Statistics

Chapter 12 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 51

Chapter 12 **Exercise 1** Page 415

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$n_1 = \text{Sample size} = 6$$
$$n_2 = \text{Sample size} = 4$$
$$\alpha = \text{Significance level} = 0.10$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 4 - 1 = 3$ and in the column with $dfn = n_1 - 1 = 6 - 1 = 5$ of the F distribution-table $\alpha/2 = 0.05$ in the appendix :

$$F_{\alpha/2} = 9.01$$

Result

$$F = 9.01$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$n_1 = \text{Sample size} = 4$$
$$n_2 = \text{Sample size} = 7$$
$$\alpha = \text{Significance level} = 0.01$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 7 - 1 = 6$ and in the column with $dfn = n_1 - 1 = 4 - 1 = 3$ of the F distribution-table $\alpha = 0.01$ in the appendix :

$$F_\alpha = 9.78$$

Result

$$F = 9.78$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$
\begin{aligned}
H_0 &: \sigma_1^2 = \sigma_2^2 \\
H_1 &: \sigma_1^2 \neq \sigma_2^2 \\
n_1 &= \text{Sample size} = 8 \\
n_2 &= \text{Sample size} = 6 \\
s_1 &= \text{Sample standard deviation} = 10 \\
s_2 &= \text{Sample standard deviation} = 7 \\
\alpha &= \text{Significance level} = 0.02
\end{aligned}
$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{10^2}{7^2} \approx 2.0408$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 6 - 1 = 5$ and in the column with $dfn = n_1 - 1 = 8 - 1 = 7$ of the F distribution-table $\alpha/2 = 0.01$ in the appendix :

$$F_{\alpha/2} = 10.5$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 2.0408 < 10.5 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the variation of the two populations.

Result

No

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$
\begin{aligned}
H_0 &: \sigma_1^2 \leq \sigma_2^2 \\
H_1 &: \sigma_1^2 > \sigma_2^2 \\
n_1 &= \text{Sample size} = 5 \\
n_2 &= \text{Sample size} = 7 \\
s_1 &= \text{Sample standard deviation} = 12 \\
s_2 &= \text{Sample standard deviation} = 7 \\
\alpha &= \text{Significance level} = 0.01
\end{aligned}
$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{12^2}{7^2} \approx 2.9388$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 7 - 1 = 6$ and in the column with $dfn = n_1 - 1 = 5 - 1 = 4$ of the F distribution-table $\alpha = 0.01$ in the appendix :

$$F_\alpha = 9.15$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 2.9388 < 9.15 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of more variation in the first population.

Result

No

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$n_2 = \text{Sample size} = 10$$
$$n_1 = \text{Sample size} = 12$$
$$s_2 = \text{Sample standard deviation} = 10$$
$$s_1 = \text{Sample standard deviation} = 12$$
$$\alpha = \text{Significance level} = 0.10$$

Given claim: Variation differs

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_a : \sigma_1^2 \neq \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{12^2}{10^2} \approx 1.44$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 10 - 1 = 9$ and in the column with $dfn = n_1 - 1 = 12 - 1 = 11$ of the F distribution-table $\alpha/2 = 0.05$ in the appendix :

$$F_{\alpha/2} = 3.10$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 1.44 < 3.10 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the variation in the listing times for men and women.

🎯 Result

No

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$n_1 = \text{Sample size} = 10$$
$$n_2 = \text{Sample size} = 8$$
$$s_1 = \text{Sample standard deviation} = 3.9$$
$$s_2 = \text{Sample standard deviation} = 3.5$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Variation is higher for oil stocks

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_a : \sigma_1^2 > \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{3.9^2}{3.5^2} \approx 1.2416$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 8 - 1 = 7$ and in the column with $dfn = n_1 - 1 = 10 - 1 = 9$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.68$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 1.2416 < 3.68 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is more variation in the oil stocks.

Result

No

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$n_1 = \text{Sample size} = 4$$
$$n_2 = \text{Sample size} = 4$$
$$n_3 = \text{Sample size} = 4$$
$$n = n_1 + n_2 + n_3 = 4 + 4 + 4 = 12$$
$$\alpha = \text{Significance level} = 0.05$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	4	4	4			12
ΣX	33	12	16			61
Mean	8.25	3	4			5.0833
ΣX^2	281	38	66			385
Variance	2.9167	0.6667	0.6667			6.8106
Std.Dev.	1.7078	0.8165	0.8165			2.6097
Std.Err.	0.8539	0.4082	0.4082			0.7534

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 12 - 3 = 9$$

The F- critical value is given in the row with $dfd = d.f._E = 9$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.26$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.26.

Step 3

(c) Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 385 - \frac{61^2}{12} \approx 74.9167$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{33^2}{44} + \frac{12^2}{4} + \frac{16^2}{4} - \frac{61^2}{12} \\ &\approx 62.1667 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 74.9167 - 62.1667 = 12.7500$$

Step 4

(d) MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{62.1667}{2} = 31.0833$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{12.7500}{9} = 1.4167$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{31.0833}{1.4167} \approx 21.94$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	74.9167	2	31.0833	21.94
Error	12.75	9	1.4167	
Total	74.9167	11		

Step 5

(e) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$21.94 > 4.26 \Rightarrow \text{ Reject } H_0$$

Result

(a) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(b) Reject H_0 when $F > 4.26$

(c) $SST = 62.1667$, $SSE = 12.75$, $SS\ Total = 74.9167$

(d)

Source	SS	df	MS	F
Treatments	74.9167	2	31.0833	21.94
Error	12.75	9	1.4167	
Total	74.9167	11		

(e) Reject H_0

Solution B

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 3 \\
n_1 &= \text{Sample size} = 4 \\
n_2 &= \text{Sample size} = 4 \\
n_3 &= \text{Sample size} = 4 \\
n &= n_1 + n_2 + n_3 = 4 + 4 + 4 = 12 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	4	4	4			12
ΣX	33	12	16			61
Mean	8.25	3	4			5.0833
ΣX^2	281	38	66			385
Variance	2.9167	0.6667	0.6667			6.8106
Std.Dev.	1.7078	0.8165	0.8165			2.6097
Std.Err.	0.8539	0.4082	0.4082			0.7534

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 12 - 3 = 9$$

The F- critical value is given in the row with $dfd = d.f._E = 9$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.26$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.26.

Step 3

(c) Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 385 - \frac{61^2}{12} \approx 74.9167$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{33^2}{44} + \frac{12^2}{4} + \frac{16^2}{4} - \frac{61^2}{12} \\ &\approx 62.1667 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 74.9167 - 62.1667 = 12.7500$$

Step 4

(d) MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{62.1667}{2} = 31.0833$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{12.7500}{9} = 1.4167$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{31.0833}{1.4167} \approx 21.94$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	74.9167	2	31.0833	21.94
Error	12.75	9	1.4167	
Total	74.9167	11		

Step 5

(e) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$21.94 > 4.26 \Rightarrow \text{ Reject } H_0$$

🎯 Result

(a) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(b) Reject H_0 when $F > 4.26$

(c) $SST = 62.1667$, $SSE = 12.75$, $SS\ Total = 74.9167$

(d)

Source	SS	df	MS	F
Treatments	74.9167	2	31.0833	21.94
Error	12.75	9	1.4167	
Total	74.9167	11		

(e) Reject H_0

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$n_1 = \text{Sample size} = 6$$
$$n_2 = \text{Sample size} = 4$$
$$n_3 = \text{Sample size} = 5$$
$$n = n_1 + n_2 + n_3 = 6 + 4 + 5 = 15$$
$$\alpha = \text{Significance level} = 0.05$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	6	4	5			15
ΣX	58	60	63			181
Mean	9.6667	15	12.6			12.0667
ΣX^2	576	934	827			2337
Variance	3.0667	11.3333	8.3			10.9238
Std.Dev.	1.7512	3.3665	2.881			3.3051
Std.Err.	0.7149	1.6833	1.2884			0.8534

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 15 - 3 = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.89$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.89.

Step 3

(c) Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 2337 - \frac{181^2}{15} \approx 152.9333$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{58^2}{6} + \frac{60^2}{4} + \frac{63^2}{5} - \frac{181^2}{15} \\ &\approx 70.4 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 152.9333 - 70.4 = 82.5333$$

Step 4

(d) MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{70.4}{2} = 35.2$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{82.5333}{12} = 6.8778$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{35.2}{6.8778} \approx 5.12$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	70.4	2	35.2	5.12
Error	82.5333	12	6.8778	
Total	152.9333	14		

Step 5

(e) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$5.12 > 3.89 \Rightarrow \text{ Reject } H_0$$

🎯 Result

(a) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(b) Reject H_0 when $F > 3.89$

(c) $SST = 70.4$, $SSE = 82.5333$, $SS\ Total = 152.9333$

(d)

Source	SS	df	MS	F
Treatments	70.4	2	35.2	5.12
Error	82.5333	12	6.8778	
Total	152.9333	14		

(e) Reject H_0

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$n_1 = \text{Sample size} = 4$$
$$n_2 = \text{Sample size} = 4$$
$$n_3 = \text{Sample size} = 4$$
$$n = n_1 + n_2 + n_3 = 4 + 4 + 4 = 12$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	4	4	4			12
ΣX	278	284	309			871
Mean	69.5	71	77.25			72.5833
ΣX^2	19396	20178	23885			63459
Variance	25	4.6667	4.9167			21.7197
Std.Dev.	5	2.1602	2.2174			4.6604
Std.Err.	2.5	1.0801	1.1087			1.3454

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 12 - 3 = 9$$

The F- critical value is given in the row with $dfd = d.f._E = 9$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.26$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.26.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 63459 - \frac{871^2}{12} \approx 238.9167$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{278^2}{4} + \frac{284^2}{4} + \frac{309^2}{4} - \frac{871^2}{12} \\ &\approx 135.1667 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 238.9167 - 135.1667 = 103.75$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{135.1667}{2} = 67.5833$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{103.75}{9} = 11.5278$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{67.5833}{11.5278} \approx 5.86$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	135.1667	2	67.5833	5.86
Error	103.75	9	11.5278	
Total	238.9167			

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$5.86 > 4.26 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean income.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 426

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$n_1 = \text{Sample size} = 5$$
$$n_2 = \text{Sample size} = 5$$
$$n_3 = \text{Sample size} = 5$$
$$n = n_1 + n_2 + n_3 = 5 + 5 + 5 = 15$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	5	5	5			15
ΣX	54	40	42			136
Mean	10.8	8	8.4			9.0667
ΣX^2	588	328	364			1280
Variance	1.2	2	2.8			3.3524
Std.Dev.	1.0954	1.4142	1.6733			1.831
Std.Err.	0.4899	0.6325	0.7483			0.4727

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 15 - 3 = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.89$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.89.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 1280 - \frac{136^2}{15} \approx 46.9333$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{54^2}{5} + \frac{40^2}{5} + \frac{42^2}{5} - \frac{136^2}{15} \\ &\approx 22.9333 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 46.9333 - 22.9333 = 24$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{22.9333}{23} = 11.4667$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{24}{12} = 2$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{11.4667}{2} \approx 5.73$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	22.9333	2	11.4667	5.73
Error	24	12	2	
Total	46.9333	14		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$5.73 > 3.89 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean number of hours spent per week by industry.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 429

Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 3 \\ n_2 &= \text{Sample size} = 5 \\ n_3 &= \text{Sample size} = 4 \\ n &= n_1 + n_2 + n_3 = 3 + 5 + 4 = 12 \\ \alpha &= \text{Significance level} = 0.05 \\ c &= \text{Confidence level} = 95\% = 0.95 \end{aligned}$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	3	5	4			12
ΣX	29	11	16			56
Mean	9.6667	2.2	4			4.6667
ΣX^2	285	27	66			378
Variance	2.3333	0.7	0.6667			10.6061
Std.Dev.	1.5275	0.8367	0.8165			3.2567
Std.Err.	0.8819	0.3742	0.4082			0.9401

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 12 - 3 = 9$$

The F- critical value is given in the row with $dfd = d.f._E = 9$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.26$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.26.

Step 3

(c) Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 378 - \frac{56^2}{12} \approx 116.6667$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{29^2}{3} + \frac{11^2}{5} + \frac{16^2}{4} - \frac{56^2}{12} \\ &\approx 107.2 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 116.6667 - 107.2 = 9.4667$$

Step 4

(d) MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{107.2}{2} = 53.6$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{9.4667}{9} = 1.0519$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{53.6}{1.0519} \approx 50.96$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	107.2	2	53.6	50.96
Error	9.4667	9	1.0519	
Total	116.6667	11		

Step 5

(e) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$50.96 > 4.26 \Rightarrow \text{ Reject } H_0$$

Step 6

(f) Determine the critical value using the Student's T distribution table in the appendix, which is given in the row $df = n - k = 12 - 3 = 9$ and in the column $\alpha/2 = (1 - c)/2 = 0.025$:

$$t_{\alpha/2} = 2.262$$

Determine the margin of error:

$$\begin{aligned} E &= t_{\alpha/2}\sqrt{MSE\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ &= 2.262\sqrt{1.0519\left(\frac{1}{3} + \frac{1}{5}\right)} \\ &\approx 1.6943 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} (\overline{x}_1 - \overline{x}_2) - E &= (9.6667 - 2.2) - 1.6943 = 7.4667 - 1.6943 = 5.7724 \\ (\overline{x}_1 - \overline{x}_2) + E &= (9.6667 - 2.2) + 1.6943 = 7.4667 + 1.6943 = 9.1610 \end{aligned}$$

Since the confidence interval $(5.7724, 9.1610)$ does not contain 0, there is sufficient evidence to conclude that treatment 1 and treatment 2 differ.

🎯 Result

(a) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(b) Reject H_0 when $F > 4.26$

(c) $SST = 107.2$, $SSE = 9.4667$, $SS\ Total = 116.6667$

(d)

Source	SS	df	MS	F
Treatments	107.2	2	53.6	50.96
Error	9.4667	9	1.0519	
Total	116.6667	11		

(e) Reject H_0

(f) Yes

PROBLEM STATEMENT

Chapter 12, Page 429

Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 6 \\ n_2 &= \text{Sample size} = 10 \\ n_3 &= \text{Sample size} = 8 \\ n &= n_1 + n_2 + n_3 = 6 + 10 + 8 = 24 \\ \alpha &= \text{Significance level} = 0.05 \\ c &= \text{Confidence level} = 95\% = 0.95 \end{aligned}$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	6	10	8			24
ΣX	15	60	34			109
Mean	2.5	6	4.25			4.5417
ΣX^2	49	384	162			595
Variance	2.3	2.6667	2.5			4.346
Std.Dev.	1.5166	1.633	1.5811			2.0847
Std.Err.	0.6191	0.5164	0.559			0.4255

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 24 - 3 = 21$$

The F- critical value is given in the row with $dfd = d.f._E = 21$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.47$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.47.

Step 3

(c) Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 595 - \frac{109^2}{24} \approx 99.9583$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{15^2}{6} + \frac{60^2}{10} + \frac{34^2}{8} - \frac{109^2}{24} \\ &\approx 46.9583 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 99.9583 - 46.9583 = 124687.99$$

Step 4

(d) MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{46.9583}{2} = 23.4792$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{53}{21} = 2.5238$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{23.4792}{2.5238} \approx 9.3$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	46.9583	2	31.0833	2.194
Error	12.75	9	1.4167	
Total	74.9167	11		

Step 5

(e) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$9.3 > 3.47 \Rightarrow \text{ Reject } H_0$$

Step 6

(f) Determine the critical value using the Student's T distribution table in the appendix, which is given in the row $df = n - k = 24 - 3 = 21$ and in the column $\alpha/2 = (1 - c)/2 = 0.025$:

$$t_{\alpha/2} = 2.080$$

Determine the margin of error:

$$\begin{aligned} E &= t_{\alpha/2}\sqrt{MSE\left(\frac{1}{n_2} + \frac{1}{n_3}\right)} \\ &= 2.080\sqrt{2.5238\left(\frac{1}{10} + \frac{1}{8}\right)} \\ &\approx 1.5674 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} (\overline{x}_2 - \overline{x}_3) - E &= (2.5 - 6) - 1.5674 = -3.5 - 1.5674 = -5.0674 \\ (\overline{x}_2 - \overline{x}_3) + E &= (2.5 - 6) + 1.5674 = -3.5 + 1.5674 = -1.9326 \end{aligned}$$

We note that the confidence interval $(-5.0674, -1.9326)$ does not contain 0, thus there is sufficient evidence to support the claim that treatment 2 and treatment 3 differ.

🎯 Result

(a) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(b) Reject H_0 when $F > 3.47$

(c) $SST = 46.9583$, $SSE = 53$, $SS\,Total = 99.9583$

(d)

Source	SS	df	MS	F
Treatments	46.9583	2	31.0833	2.194
Error	12.75	9	1.4167	
Total	74.9167	11		

(e) Reject H_0

(f) Yes

PROBLEM STATEMENT

Chapter 12, Page 429

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 4$$
$$\alpha = \text{Significance level} = 0.05$$

The P-value of the one-way ANOVA test is given in the row "Factor" and the column "P" of the given Minitab output:

$$P = 0.133$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0

$$P = 0.133 > 0.05 \Rightarrow \text{Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean number of months before a raise was granted among the four CPA firms.

Result

No

PROBLEM STATEMENT

Chapter 12, Page 429

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 4$$
$$\alpha = \text{Significance level} = 0.05$$
$$c = \text{Confidence level} = 95\% = 0.95 \quad (\text{Assumption})$$

(a) The P-value of the one-way ANOVA test is given in the row "Factor" and the column "P" of the given Minitab output:

$$P = 0.001$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0

$$P = 0.001 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean rate of return among the three types of rock.

Step 2

(b) The mean square error is given in the row "Error" and in the column "MS" of the one-way ANOVA table in the Minitab output:

$$MSE = 3.30$$

The degrees of freedom for the error is given in the row "Error" and in the column "DF" of the one-way ANOVA table in the Minitab output:

$$d.f._E = 13$$

The sample sizes of the samples are given in the column "N" of the Minitab output:

$$n_1 = 5$$
$$n_2 = 5$$
$$n_3 = 6$$

The sample means of the samples are given in the column "Mean" of the Minitab output:

$$\overline{x}_1 = 17.400$$
$$\overline{x}_2 = 11.620$$
$$\overline{x}_3 = 15.400$$

Step 3

Determine the critical value using the Student's T distribution table in the appendix, which is given in the row $df = d.f._E = 13$ and in the column $\alpha/2 = (1-c)/2 = 0.025$:

$$t_{\alpha/2} = 2.160$$

Determine the margin of error:

$$\begin{aligned} E &= t_{\alpha/2}\sqrt{MSE\left(\frac{1}{n_1}+\frac{1}{n_2}\right)} \\ &= 2.160\sqrt{3.30\left(\frac{1}{5}+\frac{1}{5}\right)} \\ &\approx 2.4817 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} (\overline{x}_1 - \overline{x}_2) - E &= (17.400 - 11.620) - 2.4817 = 5.780 - 2.4817 = 3.2983 \\ (\overline{x}_1 - \overline{x}_2) + E &= (17.400 - 11.620) + 2.4817 = 5.780 + 2.4817 = 8.2617 \end{aligned}$$

Since the confidence interval $(3.2983, 8.2617)$ does not contain 0, there is sufficient evidence to support the claim of a difference between the mean rates of return for the utility and the retail stocks.

Result

(a) Yes

(b) Yes

PROBLEM STATEMENT

Chapter 12, Page 434

Solution A

Step 1

Given:

$$k = \text{Number of treatments} = 2$$
$$b = \text{Number of blocks} = 3$$
$$n = \text{Total sample size} = 6$$
$$\alpha = \text{Significance level} = 0.05$$

(a) The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2 \text{ are equal}$$

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 2 - 1 = 1$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(2-1) = (2)(1) = 2$$

The F- critical value is given in the row with $dfd = d.f._E = 2$ and in the column with $dfn = d.f._T = 1$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 18.5$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 18.5.

Step 3

(c) The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(2-1) = (2)(1) = 2$$

The F- critical value is given in the row with $dfd = d.f._E = 2$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 19.0$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 19.0.

Step 4

(d) Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 8283 - \frac{219^2}{3 \cdot 2} \approx 289.5$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{127^2}{3} + \frac{92^2}{3} - \frac{219^2}{3 \cdot 2} \\ &\approx 204.17 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{77^2}{2} + \frac{63^2}{2} + \frac{79^2}{2} - \frac{113^2}{4 \cdot 3} \\ &\approx 76 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 289.5 - 204.17 - 76 = 9.33$$

Step 5

(e) MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{204.17}{1} = 204.17$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{9.33}{2} = 4.67$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{76}{2} = 38$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{204.17}{4.67} \approx 43.7195$$
$$F_B = \frac{MSB}{MSE} = \frac{38}{4.67} \approx 8.1370$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	204.17	1	204.17	43.7195
Blocks	76	2	38	8.1370
Error	9.33	2	4.67	
Total	289.5	5		

Step 6

(f) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 43.7195 > 18.5 \Rightarrow \text{ Reject } H_0$$
$$\textbf{Block: } 21.94 > 19.0 \Rightarrow \text{ Reject } H_0$$

🎯 Result

(a) $H_0 : \mu_1 = \mu_2$, $H_1 :$ not all of μ_1, μ_2 are equal

(b) Reject H_0 when $F > 18.5$

(c) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

Reject H_0 when $F > 19.0$

(d) $SST = 204.17$, $SSB = 76$, $SSE = 9.33$, $SS\ total = 289.5$

(e)

Source	SS	df	MS	F
Treatments	204.17	1	204.17	43.7195
Blocks	76	2	38	8.1370
Error	9.33	2	4.67	
Total	289.5	5		

(f) Reject H_0 for both sets of hypotheses

PROBLEM STATEMENT

Chapter 12, Page 435

Solution A

Step 1

Given:

$$k = \text{Number of treatments} = 3$$
$$b = \text{Number of blocks} = 3$$
$$n = \text{Total sample size} = 9$$
$$\alpha = \text{Significance level} = 0.05$$

(a) The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(3-1) = (2)(2) = 4$$

The F- critical value is given in the row with $dfd = d.f._E = 4$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 6.94$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 6.94.

Step 3

(c) The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(3-1) = (2)(2) = 4$$

The F- critical value is given in the row with $dfd = d.f._E = 4$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 6.94$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 6.94.

Step 4

(d) Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 864 - \frac{86^2}{3 \cdot 3} \approx 42.22$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{28^2}{3} + \frac{33^2}{3} + \frac{25^2}{3} - \frac{86^2}{3 \cdot 3} \\ &\approx 10.89 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{34^2}{3} + \frac{29^2}{3} + \frac{23^2}{3} - \frac{113^2}{4 \cdot 3} \\ &\approx 20.22 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 42.22 - 10.89 - 20.22 = 11.11$$

Step 5

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{10.89}{2} = 5.45$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{11.11}{4} = 2.78$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{20.22}{2} = 10.11$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{5.45}{2.78} \approx 1.96$$
$$F_B = \frac{MSB}{MSE} = \frac{10.11}{2.78} \approx 3.64$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	10.89	2	5.45	1.96
Blocks	10.89	2	5.45	3.64
Error	11.11	4	2.78	
Total	42.22	8		

Step 6

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 1.96 < 6.94 \Rightarrow \text{ Fail to reject } H_0$$
$$\textbf{Block: } 3.64 < 6.94 \Rightarrow \text{ Fail to reject } H_0$$

🎯 Result

(a) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(b) Reject H_0 when $F > 6.94$

(c) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

Reject H_0 when $F > 6.94$

(d) $SST = 10.89$, $SSB = 20.22$, $SSE = 11.11$, $SS\ total = 42.22$

(e)

Source	SS	df	MS	F
Treatments	10.89	2	5.45	1.96
Blocks	10.89	2	5.45	3.64
Error	11.11	4	2.78	
Total	42.22	8		

(f) Fail to reject H_0 for both sets of hypotheses

PROBLEM STATEMENT

Chapter 12, Page 435

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$b = \text{Number of blocks} = 5$$
$$n = \text{Total sample size} = 6$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (5-1)(3-1) = (4)(2) = 8$$

The F- critical value is given in the row with $dfd = d.f._E = 8$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.46$$

We reject the null hypothesis H_0 for the treatments when the value of the test statistic exceeds 4.46.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_5$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 5 - 1 = 4$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (5-1)(3-1) = (4)(2) = 8$$

The F- critical value is given in the row with $dfd = d.f._E = 8$ and in the column with $dfn = d.f._B = 4$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.84$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.84.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 12639 - \frac{433^2}{5 \cdot 3} \approx 139.73$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{150^2}{5} + \frac{130^2}{5} + \frac{153^2}{5} - \frac{433^2}{5 \cdot 3} \\ &\approx 62.53 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{91^2}{3} + \frac{92^2}{3} + \frac{82^2}{3} \\ &+ \frac{87^2}{3} + \frac{81^2}{3} - \frac{433^2}{5 \cdot 3} \\ &\approx 33.73 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 139.73 - 62.53 - 33.73 = 43.47$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{62.53}{2} = 31.27$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{43.47}{8} = 5.43$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{33.73}{4} = 8.43$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{31.27}{5.43} \approx 5.75$$
$$F_B = \frac{MSB}{MSE} = \frac{33.73}{5.43} \approx 1.55$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	62.53	2	31.27	5.75
Blocks	33.73	4	8.43	1.55
Error	43.47	8	5.43	
Total	139.73	14		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 5.75 > 4.46 \Rightarrow \text{ Reject } H_0$$
$$\textbf{Block: } 1.55 < 3.84 \Rightarrow \text{ Fail to reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean production rate per shift. There is not sufficient evidence to support the claim of a difference in the mean production rate per employee.

🎯 Result

Yes, by shift

PROBLEM STATEMENT

Chapter 12, Page 435

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$b = \text{Number of blocks} = 5$$
$$n = \text{Total sample size} = 6$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (5-1)(3-1) = (4)(2) = 8$$

The F- critical value is given in the row with $dfd = d.f._E = 8$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.46$$

We reject the null hypothesis H_0 for treatments when the value of the test statistic exceeds 4.46.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 5 - 1 = 4$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (5-1)(3-1) = (4)(2) = 8$$

The F- critical value is given in the row with $dfd = d.f._E = 8$ and in the column with $dfn = d.f._T = 4$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.84$$

We reject the null hypothesis H_0 for blocks when the value of the test statistic exceeds 3.84.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 6172 - \frac{298^2}{5 \cdot 3} \approx 251.73$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{88^2}{5} + \frac{112^2}{5} + \frac{98^2}{5} - \frac{298^2}{5 \cdot 3} \\ &\approx 58.13 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{56^2}{3} + \frac{58^2}{3} + \frac{52^2}{3} \\ &+ \frac{60^2}{3} + \frac{72^2}{3} - \frac{298^2}{5 \cdot 3} \\ &\approx 75.73 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 251.73 - 58.13 - 75.73 = 117.87$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{58.13}{2} = 29.07$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{117.87}{8} = 14.73$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{75.73}{4} = 18.93$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{29.07}{14.73} \approx 1.97$$
$$F_B = \frac{MSB}{MSE} = \frac{18.93}{14.73} \approx 1.28$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	58.13	2	29.07	1.97
Blocks	75.73	4	18.93	1.29
Error	117.87	8	14.73	
Total	251.73	14		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 1.97 < 4.46 \Rightarrow \text{ Fail to reject } H_0$$
$$\textbf{Block: } 1.29 < 3.84 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean number of surgeries performed by hospital. There is not sufficient evidence to support the claim of a difference in the mean number of surgeries performed by day of the week.

🎯 Result

No

PROBLEM STATEMENT

Chapter 12, Page 440

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$b = \text{Number of blocks} = 2$$
$$r = \text{Number of replications} = 3$$
$$n = \text{Total sample size} = 18$$
$$\alpha = \text{Significance level} = 0.05$$

(a) The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the total sample size n decreased by the product of the number of groups k and the number of blocks b.

$$d.f._E = n - kb = 18 - 3 \cdot 2 = 18 - 6 = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.89$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.89.

Step 2

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bkr} = 7726 - \frac{354^2}{2 \cdot 3 \cdot 3} \approx 764$$

Determine the value of the sum of squares of Factor A, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SSA &= \sum \frac{T_i^2}{br} - \frac{(\sum x)^2}{bkr} \\ &= \frac{131^2}{2 \cdot 3} + \frac{130^2}{2 \cdot 3} + \frac{93^2}{2 \cdot 3} - \frac{113^2}{2 \cdot 3 \cdot 3} \\ &\approx 156.33 \end{aligned}$$

Determine the value of the sum of squares of Factor B, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{kr} - \frac{(\sum x)^2}{bkr} \\ &= \frac{198^2}{3 \cdot 3} + \frac{156^2}{3 \cdot 3} - \frac{113^2}{2 \cdot 3 \cdot 3} \\ &\approx 98 \end{aligned}$$

Determine the value of the sum of squares of the interaction between factors A and B, where $(TB)_{ij}$ is the sum of all data values in one combination of treatment and block:

$$\begin{aligned} SSI &= \sum \frac{(TB)_{ij}^2}{r} - \frac{(\sum x)^2}{nbr} - SSA - SSB \\ &= \frac{69^2}{3} + \frac{78^2}{3} + \frac{51^2}{3} \\ &+ \frac{62^2}{3} + \frac{52^2}{3} + \frac{42^2}{3} \\ &- \frac{113^2}{2 \cdot 3 \cdot 3} - 156.33 - 98 \\ &\approx 36.34 \end{aligned}$$

Step 3

The value of the sum of squares of the error is then the value of the total-group variability decreased by the value of the sum of squares of each factor and decreased by the sum of squares of the production of the factors (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SSA + SSB + SSI + SSE$ holds):

$$SSE = \text{Total } SS - SSA - SSB - SSI = 764 - 156.33 - 98 - 36.34 = 473.33$$

MSA is SSA divided by $d.f._A$:

$$MSA = \frac{SSA}{d.f._A} = \frac{156.33}{2} = 78.18$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{98}{1} = 98$$

MSI is SSI divided by $d.f._I$:

$$MSI = \frac{SSI}{d.f._I} = \frac{36.34}{2} = 18.17$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{473.33}{12} = 39.44$$

The value of the test statistic F_A is then MSA divided by MSE, while the value F_B is MSB divided by MSE, and the value F_I is MSI divided by MSE:

$$F_A = \frac{MSA}{MSE} = \frac{78.18}{39.44} \approx 1.98$$
$$F_B = \frac{MSB}{MSE} = \frac{98}{39.44} \approx 2.48$$
$$F_I = \frac{MSI}{MSE} = \frac{18.17}{39.44} \approx 0.46$$

Step 4

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Size	156.33	2	78.17	1.98
Weight	98	1	98	2.48
Interaction	36.34	2	18.17	0.46
Error	473.33	12	39.44	
Total	764	17		

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$1.98 < 3.89 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the Size means.

Step 5

(b) The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 2 - 1 = 1$$

$d.f._E$ is the product of the total sample size n decreased by the product of the number of groups k and the number of blocks b.

$$d.f._E = n - kb = 18 - 3 \cdot 2 = 18 - 6 = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 1$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.75$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.75.

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$2.48 < 4.75 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the Weight means.

Step 6

(c) $d.f._I$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._I = (b-1)(k-1) = (2-1)(3-1) = (1)(2) = 2$$

$d.f._E$ is the product of the total sample size n decreased by the product of the number of groups k and the number of blocks b.

$$d.f._E = n - kb = 18 - 3 \cdot 2 = 18 - 6 = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.89$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.89.

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$0.46 < 3.89 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of an interaction between Weight and Size.

🎯 Result

(a) No

(b) No

(c) No

PROBLEM STATEMENT

Chapter 12, Page 441

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 4 \\
b &= \text{Number of blocks} = 3 \\
r &= \text{Number of replications} = 5 \\
n &= \text{Total sample size} = kbr = 4 \cdot 3 \cdot 5 = 60 \\
\alpha &= \text{Significance level} = 0.05 \\
SSA &= 75 \\
SSB &= 25 \\
SSI &= 300 \\
SSE &= 600
\end{aligned}
$$

Factor A

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

$d.f._E$ is the product of the total sample size n decreased by the product of the number of groups k and the number of blocks b.

$$d.f._E = n - kb = 60 - 4 \cdot 3 = 60 - 12 = 48$$

The F- critical value is given in the row with $dfd = d.f._E = 48$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.80$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.80.

Step 2

MSA is SSA divided by $d.f._A$:

$$MSA = \frac{SSA}{d.f._A} = \frac{75}{3} = 25$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{25}{2} = 12.5$$

MSI is SSI divided by $d.f._I$:

$$MSI = \frac{SSI}{d.f._I} = \frac{300}{6} = 50$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{600}{48} = 12.5$$

The value of the test statistic F_A is then MSA divided by MSE, while the value F_B is MSB divided by MSE, and the value F_I is MSI divided by MSE:

$$F_A = \frac{MSA}{MSE} = \frac{25}{12.5} \approx 2$$
$$F_B = \frac{MSB}{MSE} = \frac{12.5}{12.5} \approx 1$$
$$F_I = \frac{MSI}{MSE} = \frac{50}{12.5} \approx 4$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Factor A	75	3	25	2
Factor B	25	2	12.5	1
Interaction	300	6	50	4
Error	600	48	12.5	
Total	1000	59		

Step 3

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$2 < 2.80 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the Factor A means.

Step 4

Factor B

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the total sample size n decreased by the product of the number of groups k and the number of blocks b.

$$d.f._E = n - kb = 60 - 4 \cdot 3 = 60 - 12 = 48$$

The F- critical value is given in the row with $dfd = d.f._E = 48$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.19$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.19.

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$1 < 3.19 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the Factor B means.

Step 5

Interaction

$d.f._I$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._I = (b-1)(k-1) = (4-1)(3-1) = (3)(2) = 6$$

$d.f._E$ is the product of the total sample size n decreased by the product of the number of groups k and the number of blocks b.

$$d.f._E = n - kb = 60 - 4 \cdot 3 = 60 - 12 = 48$$

The F- critical value is given in the row with $dfd = d.f._E = 48$ and in the column with $dfn = d.f._T = 9$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.29$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.29.

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$4 > 2.29 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of an interaction between Factor A and Factor B.

🎯 Result

There is not sufficient evidence to support the claim of a difference in the Factor A means. There is not sufficient evidence to support the claim of a difference in the Factor B means. There is sufficient evidence to support the claim of an interaction between Factor A and Factor B.

Exercise 21

PROBLEM STATEMENT

Chapter 12, Page 441

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 3 \\
b &= \text{Number of blocks} = 2 \\
r &= \text{Number of replications} = 4 \\
n &= \text{Total sample size} = 3 \times 2 \times 4 = 24 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

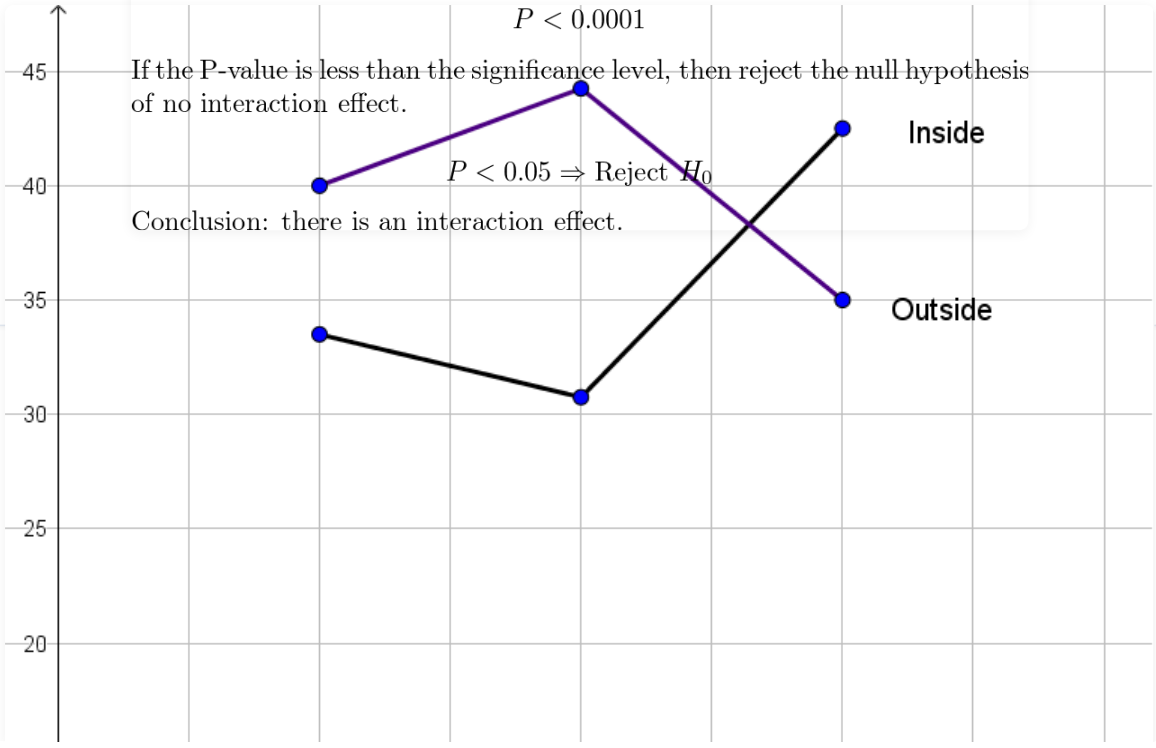
(a) Let us first determine the mean of all six combinations of machine and position. The mean is the sum of all data values divided by the number of data values.

$$
\begin{aligned}
\overline{x}_{11} &= \frac{\sum x_i}{n} = \frac{33+40+30+31}{4} = 33.5 \\
\overline{x}_{12} &= \frac{\sum x_i}{n} = \frac{29+28+33+33}{4} = 30.75 \\
\overline{x}_{13} &= \frac{\sum x_i}{n} = \frac{47+39+39+45}{4} = 42.5 \\
\overline{x}_{21} &= \frac{\sum x_i}{n} = \frac{43+36+41+40}{4} = 40 \\
\overline{x}_{22} &= \frac{\sum x_i}{n} = \frac{48+45+40+44}{4} = 44.25 \\
\overline{x}_{23} &= \frac{\sum x_i}{n} = \frac{37+32+36+35}{4} = 35
\end{aligned}
$$

Next, we create the interaction graph. The machine is on the horizontal axis, while the mean is on the vertical axis. We connect consecutive points corresponding to the same positions with a straight line.

We note that the lines intersect in the interaction graph below, which indicates that there is an interaction effect.

Moreover, the mean sales appear to be highest for d-320 when the position is outside, while the mean sales are then lowest for uv-57. The mean sales appear to be highest for uv-57 when the position is inside, while the mean sales are then lowest for d-320.



Step 2

(b) I used technology to determine the following two-way ANOVA table (for example, the website vassarstats.net contains a free two-way ANOVA calculator that will calculate the ANOVA table below).

Source	df	SS	MS	F	P
Rows	1	104.17	104.17	9.12	0.0074
Columns	2	16.33	8.16	0.72	0.5002
Interaction	2	454.33	227.17	19.89	<0.0001
Error	18	205.5	11.42		
Total	23	780.33			

Interaction effect

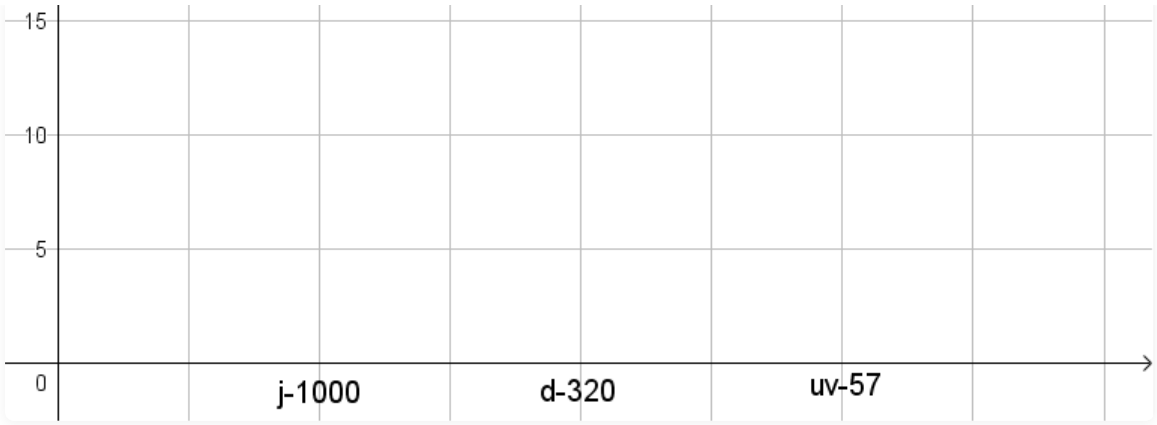
The P-value of the interaction effect is given in the output in the row with Interaction and in the column with P:

$$P < 0.0001$$

If the P-value is less than the significance level, then reject the null hypothesis of no interaction effect.

$$P < 0.05 \Rightarrow \text{Reject } H_0$$

Conclusion: there is an interaction effect.



Step 3

Position

The P-value for the row variable is given in the output in the row with ”Rows” and in the column with P:

$$P = 0.0075$$

If the P-value is less than the significance level, then reject the null hypothesis.

$$0.0075 < 0.05 \Rightarrow \text{Reject } H_0$$

Conclusion: There is a significant effect from the row factor "Position".

Machine

The P-value for the row variable is given in the output in the row with ”Columns” and in the column with P:

$$P = 0.5002$$

If the P-value is less than the significance level, then reject the null hypothesis.

$$0.5002 > 0.05 \Rightarrow \text{Fail to reject } H_0$$

Conclusion: There is no significant effect from the column factor "Machine".

Step 4

(c) We need to compare the inside and outside mean sales for each machine, which is done by constructing a one-way ANOVA table per machine (for example, the website vassarstats.net contains a free one-way ANOVA calculator that will calculate the ANOVA tables below).

J-1000

Source	df	SS	MS	F	P
Position	1	84.5	84.5	5.83	0.052
Error	6	87.0	14.5		
Total	7	171.5			

D-320

Source	df	SS	MS	F	P
Position	1	364.50	364.50	40.88	0.001
Error	6	53.50	8.92		
Total	7	418.00			

UV-57

Source	df	SS	MS	F	P
Position	1	112.5	112.5	10.38	0.018
Error	6	65.0	10.8		
Total	7	177.5			

Step 5

If the P-value is less than the significance level, then reject the null hypothesis of an effect of Position on the mean sales.

$$0.001 < 0.05 \Rightarrow \text{Reject } H_0$$
$$0.052 > 0.05 \Rightarrow \text{Fail to reject } H_0$$
$$0.018 < 0.05 \Rightarrow \text{Reject } H_0$$

Conclusion: There is no significant effect of Position on the machine J-1000, while there is an effect of Position on D-320 and UV-57.

We then recommend using the position outside for the D-320 machine and the position inside for the UV-57 (as these positions have the highest mean sales), while the position doesn't matter for the machine J-500.

🎯 Result

(a) Yes

(b) Significant interaction effect, Significant effect from Position, No significant effect of machine.

(c) We then recommend using the position outside for the D-320 machine and the position inside for the UV-57, while the position doesn't matter for the machine J-500.

PROBLEM STATEMENT

Chapter 12, Page 441

Solution A

Step 1

For this entire problem, we will be working on software with statistical tools. You may also use other application that has one-way ANOVA and two-way ANOVA test because these are the test that we are going to perform. We can simply encode the data on the software then use its features to create an interaction graph. In addition, we will be using the encoded data to compute a two-way ANOVA on the software. For the last question, we will perform a one-way ANOVA on each area group and use the results to make recommendations.

Step 2

(a) Interaction Graph:

Here, we can use software to easily find the means of gender and areas and automatically graph the obtained values. To do this, let us create a table of the data on the worksheet:

	Area A	Area B	Area C
Female	1016	1045	770
	1007	845	733
	875	848	844
	968	904	771
Male	978	1154	926
	1056	1091	1055
	982	878	1066
	918	876	1088

Is there an interaction effect?

By just looking at the obtained graph, we can say that there is an interaction effect of work area and gender on the salary because the two lines are not parallel with each other. Here, the factors, work area, and gender follow different patterns or trends which indicates that there has been an interaction effect on the salary.

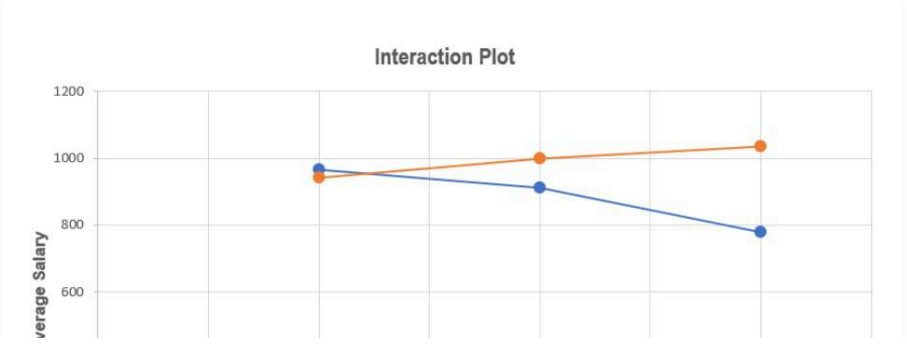
Describing their Interaction

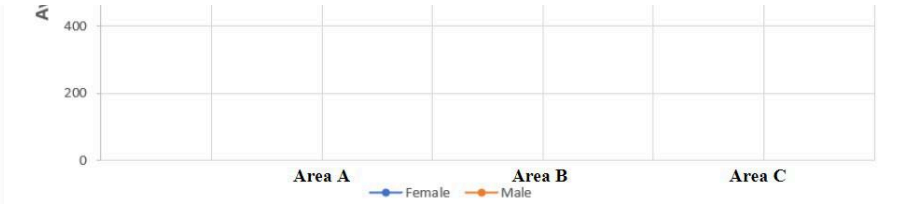
The salary will depend on the combined effects of the gender and the working area. By observing the trend of the average salaries, on area A male has greater average salary than female while on the areas B and C, female has greater average salary than male. There will be many external factors that may affect the interactions between gender and working area on the salary but the question that we may raise is that if the interactions are significant or the differences are true.

Using the *average* formula in the software, we can find the mean of each area per gender. The mean salary per gender in each work area are:

	Area A	Area B	Area C
Female	966.5	910.5	779.5
Male	941	999.75	1033.75

Now, highlight the table for the averages then proceed to insert it on the menu bar. There you can find charts that automatically graph the highlighted data. Here we will be using the *scatter with straight-line* chart then modifying its labels and titles. We let the horizontal axis be the work area while the vertical axis is the average salary.





Step 4

ANOVA:

Hypothesis:

We want to test for gender, area, and interaction effects on salary. Thus, we test the hypotheses:

1. H_0: The gender means are the same.
   H_1: The gender means are not the same.
2. H_0: The work area means are the same.
   H_1: The work area means are not the same.
3. H_0: There is no interaction between gender and work area.
   H_1: There is an interaction between gender and work area.

Decision Rule

The significance level is 0.05 thus our decision rule to all the hypotheses is:

$$\text{Reject } H_0 \longrightarrow p - \text{values} < 0.05$$

Step 5

Here, we are going to perform *two-way ANOVA with interactions*. We may use other statistical applications on the internet but here let us use the software's statistical tools add-ins. To do this, use the data table we input in part (a). Go to **Data** menu then find **data analysis** toolbar. A dialog box will appear. We should choose **ANOVA:Two-Factor with Replication** then on the *input range*, highlight the cells with the data and include their titles. Type 4 on the *rows per sample* because there are 4 observations in each gender. *Alpha* should be 0.05 and the *output range* should be the cell where we want to put the results. By doing these steps we will get the results for the ANOVA test:

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Sample	67416	1	67416	6.851819	0.017441	4.413873
Columns	12199.75	2	6099.875	0.61996	0.549064	3.554557
Interaction	79101.75	2	39550.88	4.01975	0.036038	3.554557
Within	177104.5	18	9839.139			
Total	335822	23				

The sample row corresponds to gender, columns row corresponds to the work area. Now, let us first interpret the results on gender. The computed $p-$value is 0.0174. From the decision rule, we will reject the null hypothesis and conclude that the **gender means are not the same**. Then for the work areas, the computed $p-$value is 0.5490. Since this is greater than 0.05 then, we fail to reject the null hypothesis. Thus, **the work area means are the same**. Finally, the $p-$value of the interaction is 0.0360 which is less than the significance level 0.05. Reject the null and conclude that **there is an interaction between gender and area**.

Step 6

(c) Gender Mean Sales for each Work Area:

We want to know whether the means of two independent groups, female and male, have a significant difference for each work area. We need to conduct one-way ANOVA for each work area. We can still use statistical tools in the software to find the $p-$values. We will be working with these hypotheses:

Testing for Significant Differences between Gender Means in Area A

H_0: There is no significant difference between gender means.

H_1: There is a significant difference between gender means.

Testing for Significant Differences between Gender Means in Area B

H_0: There is no significant difference between gender means.

H_1: There is a significant difference between gender means.

Testing for Significant Differences between Gender Means in Area C

H_0: There is no significant difference between gender means.
H_1: There is a significant difference between gender means.

Decision Rule

We will reject the null hypothesis if the computed $p-$value is less than the significant level 0.05.

$$\text{Reject } H_0 \longrightarrow p - \text{values} < 0.05$$

Step 7

On the worksheet, create a table where salaries per gender are arranged in columns per work area:

Area A:

Female	Male
1016	978
1007	1056
875	982
968	748

Area B:

Female	Male
1045	1154
845	1091
848	878
904	876

Area C:

Female	Male
770	926
733	1055
844	1066
771	1088

Step 8

Click on **Data** on the menu bar then **Data Analysis**. Choose **ANOVA: Single Factor** then OK, highlight the two columns with the labels, choose **Grouped By: Columns**, check the **Labels in first row** box, set the **Aplha:0.05**, and choose where to put the results on the **Output Range**. Finally, click OK. Do this on the three groups of data. The results are:

Gender Means in Area A

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	1300.5	1	1300.5	0.118247	0.742674	5.987378
Within Groups	65989	6	10998.16667			
Total	67289.5	7				

Gender Means in Area B

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	15931.13	1	15931.13	1.079031	0.338952	5.987378
Within Groups	88585.75	6	14764.29			
Total	104516.9	7				

Gender Means in Area C

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	129286.1	1	129286.1	34.43077	0.001084	5.987378
Within Groups	22529.75	6	3754.958			
Total	151815.9	7				

Step 9

Decision:

For Area A, the computed $p-$value is 0.7427. Since it is greater than 0.05 then we **fail to reject the null hypothesis**. Thus, there is no significant difference between gender means on area A.

For Area B, the computed $p-$value is 0.3390. Since it is greater than 0.05 then we **fail to reject the null hypothesis**. Thus, there is no significant difference between gender means on area B.

For Area C, the computed $p-$value is 0.0011. Since it is less than 0.05 then we will **reject the null hypothesis**. Thus, there is a significant difference between gender means on area C.

Step 10

Recommendation:

Here, we can say that the company pays both female and male employees equally in areas A and B but they differ on area C. Looking at the data on Area C, females have lower salaries recorded than males. This is why employees are claiming that the company pays females less than men. We would like to recommend increasing the salaries for females or decreasing salaries for males in order to have their means equal and that there will be no complaints about unequal pay.

💡 Solution B

Step 1

For this entire problem, we will be working on software with statistical tools. You may also use other application that has one-way ANOVA and two-way ANOVA test because these are the test that we are going to perform. We can simply encode the data on the software then use its features to create an interaction graph. In addition, we will be using the encoded data to compute a two-way ANOVA on the software. For the last question, we will perform a one-way ANOVA on each area group and use the results to make recommendations.

Step 2

(a) Interaction Graph:

Here, we can use software to easily find the means of gender and areas and automatically graph the obtained values. To do this, let us create a table of the data on the worksheet:

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Is there an interaction effect?

By just looking at the obtained graph, we can say that there is an interaction effect of work area and gender on the salary because the two lines are not parallel with each other. Here, the factors, work area, and gender follow different patterns or trends which indicates that there has been an interaction effect on the salary.

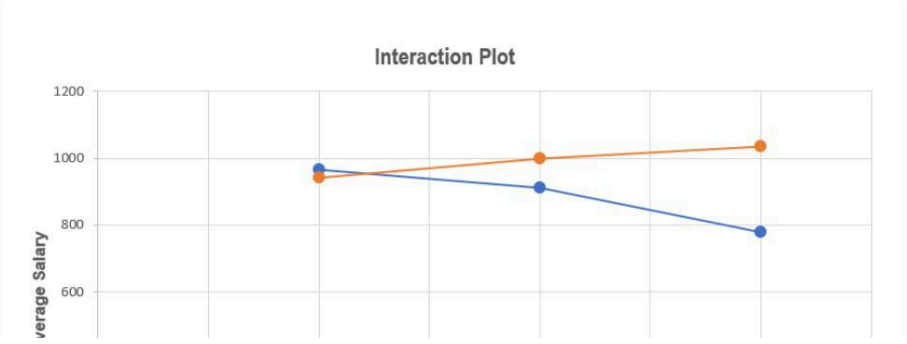
Describing their Interaction

The salary will depend on the combined effects of the gender and the working area. By observing the trend of the average salaries, on area A male has greater average salary than female while on the areas B and C, female has greater average salary than male. There will be many external factors that may affect the interactions between gender and working area on the salary but the question that we may raise is that if the interactions are significant or the differences are true.

Using the *average* formula in the software, we can find the mean of each area per gender. The mean salary per gender in each work area are:

	Area A	Area B	Area C
Female	966.5	910.5	779.5
Male	941	999.75	1033.75

Now, highlight the table for the averages then proceed to insert it on the menu bar. There you can find charts that automatically graph the highlighted data. Here we will be using the *scatter with straight-line* chart then modifying its labels and titles. We let the horizontal axis be the work area while the vertical axis is the average salary.





Step 4

ANOVA:

Hypothesis:

We want to test for gender, area, and interaction effects on salary. Thus, we test the hypotheses:

1. H_0: The gender means are the same.
   H_1: The gender means are not the same.
2. H_0: The work area means are the same.
   H_1: The work area means are not the same.
3. H_0: There is no interaction between gender and work area.
   H_1: There is an interaction between gender and work area.

Decision Rule

The significance level is 0.05 thus our decision rule to all the hypotheses is:

$$\text{Reject } H_0 \longrightarrow p - \text{values} < 0.05$$

Step 5

Here, we are going to perform *two-way ANOVA with interactions*. We may use other statistical applications on the internet but here let us use the software's statistical tools add-ins. To do this, use the data table we input in part (a). Go to **Data** menu then find **data analysis** toolbar. A dialog box will appear. We should choose **ANOVA:Two-Factor with Replication** then on the *input range*, highlight the cells with the data and include their titles. Type 4 on the *rows per sample* because there are 4 observations in each gender. *Alpha* should be 0.05 and the *output range* should be the cell where we want to put the results. By doing these steps we will get the results for the ANOVA test:

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
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Interaction	79101.75	2	39550.88	4.01975	0.036038	3.554557
Within	177104.5	18	9839.139			
Total	335822	23				

The sample row corresponds to gender, columns row corresponds to the work area. Now, let us first interpret the results on gender. The computed $p-$value is 0.0174. From the decision rule, we will reject the null hypothesis and conclude that the **gender means are not the same**. Then for the work areas, the computed $p-$value is 0.5490. Since this is greater than 0.05 then, we fail to reject the null hypothesis. Thus, **the work area means are the same**. Finally, the $p-$value of the interaction is 0.0360 which is less than the significance level 0.05. Reject the null and conclude that **there is an interaction between gender and area**.

Step 6

(c) Gender Mean Sales for each Work Area:

We want to know whether the means of two independent groups, female and male, have a significant difference for each work area. We need to conduct one-way ANOVA for each work area. We can still use statistical tools in the software to find the $p-$values. We will be working with these hypotheses:

Testing for Significant Differences between Gender Means in Area A

H_0: There is no significant difference between gender means.

H_1: There is a significant difference between gender means.

Testing for Significant Differences between Gender Means in Area B

H_0: There is no significant difference between gender means.

H_1: There is a significant difference between gender means.

Testing for Significant Differences between Gender Means in Area C

H_0: There is no significant difference between gender means.
H_1: There is a significant difference between gender means.

Decision Rule

We will reject the null hypothesis if the computed $p-$value is less than the significant level 0.05.

$$\text{Reject } H_0 \longrightarrow p - \text{values} < 0.05$$

Step 7

On the worksheet, create a table where salaries per gender are arranged in columns per work area:

Area A:

Female	Male
1016	978
1007	1056
875	982
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Area B:

Female	Male
1045	1154
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904	876

Area C:

Female	Male
770	926
733	1055
844	1066
771	1088

Step 8

Click on **Data** on the menu bar then **Data Analysis**. Choose **ANOVA: Single Factor** then OK, highlight the two columns with the labels, choose **Grouped By: Columns**, check the **Labels in first row** box, set the **Aplha:0.05**, and choose where to put the results on the **Output Range**. Finally, click OK. Do this on the three groups of data. The results are:

Gender Means in Area A

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	1300.5	1	1300.5	0.118247	0.742674	5.987378
Within Groups	65989	6	10998.16667			
Total	67289.5	7				

Gender Means in Area B

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	15931.13	1	15931.13	1.079031	0.338952	5.987378
Within Groups	88585.75	6	14764.29			
Total	104516.9	7				

Gender Means in Area C

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	129286.1	1	129286.1	34.43077	0.001084	5.987378
Within Groups	22529.75	6	3754.958			
Total	151815.9	7				

Step 9

Decision:

For Area A, the computed $p-$value is 0.7427. Since it is greater than 0.05 then we **fail to reject the null hypothesis**. Thus, there is no significant difference between gender means on area A.

For Area B, the computed $p-$value is 0.3390. Since it is greater than 0.05 then we **fail to reject the null hypothesis**. Thus, there is no significant difference between gender means on area B.

For Area C, the computed $p-$value is 0.0011. Since it is less than 0.05 then we will **reject the null hypothesis**. Thus, there is a significant difference between gender means on area C.

Step 10

Recommendation:

Here, we can say that the company pays both female and male employees equally in areas A and B but they differ on area C. Looking at the data on Area C, females have lower salaries recorded than males. This is why employees are claiming that the company pays females less than men. We would like to recommend increasing the salaries for females or decreasing salaries for males in order to have their means equal and that there will be no complaints about unequal pay.

PROBLEM STATEMENT

Chapter 12, Page 443

Solution A

Step 1

Given:

$$n_1 = \text{Sample size} = 21$$
$$n_2 = \text{Sample size} = 18$$
$$s_1 = \text{Sample standard deviation} = 45600$$
$$s_2 = \text{Sample standard deviation} = 21330$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: More variation on oceanfront homes

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 \leq \sigma_2^2$$
$$H_a : \sigma_1^2 > \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{45600^2}{21330^2} \approx 4.5703$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 18 - 1 = 17$ and in the column with $dfn = n_1 - 1 = 21 - 1 = 20$ of the F distribution-table $\alpha = 0.01$ in the appendix :

$$F_\alpha = 3.16$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 4.5703 > 3.16 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is more variation in the selling prices of the oceanfront homes.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 443

Solution A

Step 1

Given:

$$n_1 = \text{Sample size} = 16$$
$$n_2 = \text{Sample size} = 16$$
$$s_1 = \text{Sample standard deviation} = 22$$
$$s_2 = \text{Sample standard deviation} = 12$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: More variation for new machine

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 \leq \sigma_2^2$$
$$H_1 : \sigma_1^2 > \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{22^2}{12^2} \approx 3.3611$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 16 - 1 = 15$ and in the column with $dfn = n_1 - 1 = 16 - 1 = 15$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_{\alpha/2} = 2.40$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 3.3611 > 2.40 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is more variation in the processing time of the new machine.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 444

Solution A

Step 1

Given:

$$n_2 = \text{Sample size} = 7$$
$$n_1 = \text{Sample size} = 8$$
$$\alpha = \text{Significance level} = 0.01$$

Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\overline{x}_2 = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{98 + 78 + ... + 70}{7} = 69.8571$$
$$\overline{x}_1 = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{75 + 81 + ... + 101}{8} = 69.25$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(98 - 69.8571)^2 + ... + (70 - 69.8571)^2}{7 - 1}} \\ &= 14.7922 \end{aligned}$$

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(75 - 69.25)^2 + ... + (101 - 69.25)^2}{8 - 1}} \\ &= 22.9518 \end{aligned}$$

Step 2

Hypothesis test

Given claim: Variation is lower for Sharkey

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 \leq \sigma_2^2$$
$$H_1 : \sigma_1^2 > \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{22.9518^2}{14.7922^2} \approx 2.4075$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 7 - 1 = 6$ and in the column with $dfn = n_1 - 1 = 8 - 1 = 7$ of the F distribution-table $\alpha = 0.01$ in the appendix :

$$F_{\alpha/2} = 8.26$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 2.4075 < 8.26 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that Sharkey's sales are more consistent and thus we do not agree with Mr. Sharkey.

Result

No

PROBLEM STATEMENT

Chapter 12, Page 444

Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 5 \\ n_2 &= \text{Sample size} = 5 \\ n_3 &= \text{Sample size} = 5 \\ n &= n_1 + n_2 + n_3 = 5 + 5 + 5 = 15 \\ \alpha &= \text{Significance level} = 0.05 \\ SS_{total} &= 100 \\ SST &= 40 \end{aligned}$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 15 - 3 = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.89$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.89.

Step 3

(c) The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 100 - 40 = 60$$

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{40}{2} = 20$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{60}{12} = 5$$

The value of the test statistic F is then MST divided by MSE:

$$F = \frac{MST}{MSE} = \frac{20}{5} \approx 4$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	40	2	20	4
Error	60	12	5	
Total	100	14		

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$4 > 3.89 \Rightarrow \text{ Reject } H_0$$

🎯 **Result**

(a) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(b) Reject H_0 when $F > 3.89$

(c) $F = 4$

(d) Reject H_0

PROBLEM STATEMENT

Chapter 12, Page 444

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 4 \\
n_1 &= \text{Sample size} = 6 \\
n_2 &= \text{Sample size} = 6 \\
n_3 &= \text{Sample size} = 6 \\
n_4 &= \text{Sample size} = 6 \\
n &= n_1 + n_2 + n_3 + n_4 = 6 + 6 + 6 + 6 = 24 \\
\alpha &= \text{Significance level} = 0.05 \\
SS_{total} &= 250 \\
MSE &= 10
\end{aligned}
$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

Step 2

(b) $d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 24 - 4 = 20$$

The F- critical value is given in the row with $dfd = d.f._E = 20$ and in the column with $dfn = d.f._T = 3$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.10$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.10.

Step 3

(c) MSE is SSE divided by $d.f._E$, thus SSE is the product of MSE and $d.f._E$:

$$SSE = MSE \times d.f._E = 10 \times 20 = 200$$

Since thee sum $SS_{total} = SST + SSE$ holds:

$$SST = SS_{total} - SSE = 250 - 200 = 50$$

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{50}{3} = 16.6667$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{200}{20} = 10$$

The value of the test statistic F is then MST divided by MSE:

$$F = \frac{MST}{MSE} = \frac{16.6667}{10} \approx 1.67$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	50	3	16.6667	1.67
Error	200	20	10	
Total	250	23		

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$1.67 < 3.10 \Rightarrow \text{ Fail to reject } H_0$$

🎯 **Result**

(a) $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$, $H_1 :$ not all of $\mu_1, \mu_2, \mu_3, \mu_4$ are equal

(b) Reject H_0 when $F > 3.10$

(c) $F = 1.67$

(d) Fail to reject H_0

PROBLEM STATEMENT

Chapter 12, Page 444

Solution A

Step 1

Given:

$$\begin{aligned} \alpha &= \text{Significance level} = 0.05 \\ d.f._T &= 2 \\ MSE &= 20 \\ SS_{total} &= 500 \\ d.f._{total} &= 11 \end{aligned}$$

(a) The degrees of freedom for treatments is the number of treatments decreased by 1, which implies that the number of treatments is the degrees of freedom for treatments increased by 1.

$$\text{Number of treatments} = d.f._T + 1$$

Step 2

(b) The total sample size is the total degrees of freedom increased by 1 (as the total degrees of freedom is the total sample size decreased by 1):

$$n = d.f._{total} + 1 = 11 + 1 = 12$$

Step 3

(c) The missing degrees of freedom is obtain by decreasing the total degrees of freedom by the other known degrees of freedom.

$$d.f._E = d.f._{total} - d.f._T = 11 - 2 = 9$$

The F- critical value is given in the row with $dfd = d.f._E = 9$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.26$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.26.

Step 4

(d) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Step 5

(e) MSE is SSE divided by $d.f._E$, which implies that SSE is the product of MSE and $d.f._E$.

$$SSE = MSE \times d.f._E = 20 \times 9 = 180$$

Since the sum $SS_{total} = SST + SSE$ holds:

$$SST = SS_{total} - SSE = 500 - 180 = 320$$

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{320}{2} = 160$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{160}{20} = 8$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	320	2	160	8
Error	180	9	20	
Total	500	11		

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$8 > 4.26 \Rightarrow \text{ Reject } H_0$$

Result

(a) 3 treatments

(b) $n = 12$

(c) $F = 4.26$

(d) $H_0 : \mu_1 = \mu_2 = \mu_3$, $H_1 :$ not all of μ_1, μ_2, μ_3 are equal

(e) Reject H_0

Chapter 12 **Exercise 29** Page 444

PROBLEM STATEMENT

Chapter 12, Page 444

Solution A

Step 1

	SS	df	MS	F	P
Treatment	63.3333	2	31.6667	13.38	0.000881
Error	28.4	12	2.3667		
Total	91.7333	14			

We note that the P-value 0.000881 is smaller than the significance level 0.05 and thus there is sufficient evidence to support the claim of a difference in the price of a particular toy at three different types of stores.

Result

There is sufficient evidence to support the claim of a difference in the price of a particular toy at three different types of stores.

PROBLEM STATEMENT

Chapter 12, Page 444

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 3 \\
n_1 &= \text{Sample size} = 14 \\
n_2 &= \text{Sample size} = 14 \\
n_3 &= \text{Sample size} = 7 \\
n &= n_1 + n_2 + n_3 = 14 + 14 + 7 = 35 \\
\alpha &= \text{Significance level} = 0.05 \\
c &= \text{Confidence level} = 95\% = 0.95
\end{aligned}
$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	14	14	7			35
ΣX	707	729	390			1826
Mean	50.5	52.0714	55.7143			52.1714
ΣX^2	35963	38151	21948			96062
Variance	19.9615	14.6868	36.5714			23.4403
Std.Dev.	4.4678	3.8323	6.0474			4.8415
Std.Err.	1.1941	1.0242	2.2857			0.8184

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 35 - 3 = 32$$

The F- critical value is given in the row with $dfd = d.f._E = 32$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.29$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.29.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 96062 - \frac{1826^2}{35} \approx 796.9714$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{707^2}{14} + \frac{729^2}{14} + \frac{390^2}{7} - \frac{1826^2}{35} \\ &\approx 127.1143 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 796.9714 - 127.1143 = 669.8571$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{127.1143}{2} = 63.5571$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{669.8571}{32} = 20.9330$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{63.5571}{20.9330} \approx 3.04$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	127.1143	2	63.5571	3.04
Error	669.8571	32	20.9330	
Total	796.9714	34		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$3.04 < 3.29 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean flight times among the three airlines.

Step 6

(b) Determine the critical value using the Student's T distribution table in the appendix, which is given in the row $df = n - k = 35 - 3 = 32$ and in the column $\alpha/2 = (1 - c)/2 = 0.025$:

$$t_{\alpha/2} = 2.037$$

Determine the margin of error:

$$\begin{aligned} E &= t_{\alpha/2}\sqrt{MSE\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ &= 2.037\sqrt{20.9330\left(\frac{1}{14} + \frac{1}{7}\right)} \\ &\approx 4.3142 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} (\overline{x}_1 - \overline{x}_2) - E &= (50.5 - 55.7143) - 4.3142 = -5.2143 - 4.3142 = -9.5285 \\ (\overline{x}_1 - \overline{x}_2) + E &= (50.5 - 55.7143) + 4.3142 = -5.2143 + 4.3142 = -0.9001 \end{aligned}$$

Thus we 90% confidence interval for the difference in the means between Goust and Cloudtran is $(-9.5285, -0.9001)$.

Result

(a) No

(b) $(-9.5285, -0.9001)$

PROBLEM STATEMENT

Chapter 12, Page 445

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 4 \\
n_1 &= \text{Sample size} = 6 \\
n_2 &= \text{Sample size} = 6 \\
n_3 &= \text{Sample size} = 6 \\
n_4 &= \text{Sample size} = 6 \\
n &= n_1 + n_2 + n_3 + n_4 = 6 + 6 + 6 + 6 = 24 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	6	6	6	6		24
ΣX	86	108	81	104		379
Mean	14.3333	18	13.5	17.3333		15.7917
ΣX^2	1236	1980	1103	1818		6137
Variance	0.6667	7.2	1.9	3.0667		6.6069
Std.Dev.	0.8165	2.6833	1.3784	1.7512		2.5704
Std.Err.	0.3333	1.0954	0.5627	0.7149		0.5247

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 24 - 4 = 20$$

The F- critical value is given in the row with $dfd = d.f._E = 20$ and in the column with $dfn = d.f._T = 3$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.10$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.10.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 6137 - \frac{379^2}{24} \approx 151.9583$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{86^2}{6} + \frac{108^2}{6} + \frac{81^2}{6} + \frac{104^2}{6} - \frac{379^2}{24} \\ &\approx 87.7917 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 151.9583 - 87.7917 = 64.1667$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{87.7917}{3} = 29.2639$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{64.1667}{20} = 3.2083$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{29.2639}{3.2083} \approx 9.12$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	87.7917	3	29.2639	9.12
Error	64.1667	20	3.2083	
Total	151.9583	23		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$9.12 > 3.10 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean number of crimes.

Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 445

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 3 \\
n_1 &= \text{Sample size} = 9 \\
n_2 &= \text{Sample size} = 12 \\
n_3 &= \text{Sample size} = 11 \\
n &= n_1 + n_2 + n_3 = 9 + 12 + 11 = 32 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	9	12	11			32
ΣX	252	557	579			1388
Mean	28	46.4167	52.6364			43.375
ΣX^2	7586	26275	30823			64684
Variance	66.25	38.2652	34.6545			144.5
Std.Dev.	8.1394	6.1859	5.8868			12.0208
Std.Err.	2.7131	1.7857	1.7749			2.125

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 32 - 3 = 29$$

The F- critical value is given in the row with $dfd = d.f._E = 29$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.33$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.33.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 64684 - \frac{1388^2}{32} \approx 4479.5$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{252^2}{9} + \frac{557^2}{12} + \frac{579^2}{11} - \frac{1388^2}{32} \\ &\approx 3182.0379 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 4479.5 - 3182.0379 = 1297.4621$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{3182.0379}{2} = 1591.0189$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{1297.4621}{29} = 44.7401$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{1591.0189}{44.7401} \approx 35.56$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	3182.0379	2	1591.0189	35.56
Error	1297.4621	29	44.7401	
Total	4479.5	31		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$35.56 > 3.33 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean attention span of the children for the various commercials.

This then also implies that one of the pairs of means needs to be significantly different.

Since the sample of clothes has a much lower mean than the other two samples, we would recommend dropping "clothes."

🎯 Result

Yes, Yes

PROBLEM STATEMENT

Chapter 12, Page 445

Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of groups} = 2 \\ n_1 &= \text{Sample size} = 6 \\ n_2 &= \text{Sample size} = 8 \\ n &= n_1 + n_2 = 6 + 8 = 14 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	6	8				14
ΣX	114	216				330
Mean	19	27				23.5714
ΣX^2	2208	5904				8112
Variance	8.4	10.2857				25.6484
Std.Dev.	2.8983	3.2071				5.0644
Std.Err.	1.1832	1.1339				1.3535

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 2 - 1 = 1$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 14 - 2 = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 1$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.75$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.75.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 8112 - \frac{330^2}{144} \approx 333.4286$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{114^2}{6} + \frac{216^2}{8} - \frac{330^2}{14} \\ &\approx 219.4286 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 333.4286 - 219.4286 = 114$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{219.4286}{1} = 219.4286$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{114}{12} = 9.5$$

The value of the test statistic F is then MST divided by MSE:

$$F = \frac{MST}{MSE} = \frac{219.4286}{9.5} \approx 23.1$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	219.4286	1	219.4286	23.1
Error	114	12	9.5	
Total	333.4286	13		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$23.1 > 4.75 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim that the population means are equal.

Step 6

(b) Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 6 + 8 - 2 = 12$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.179$$

The rejection region then contains all values smaller than -2.179 and all values greater than 2.179,thus we reject H_0 when $t < -2.179$ or $t > 2.179$.

Step 7

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(6 - 1)2.8983^2 + (8 - 1)3.2071^2}{6 + 8 - 2}} \approx 3.0822$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{19 - 27}{3.0822\sqrt{\frac{1}{6} + \frac{1}{8}}} \approx -4.8060$$

Step 8

(c) By part (a): $F = 23.1$

By part (b): $t = -4.8060$

We then note that the equality $t^2 = F$ indeed holds:

$$t^2 = (-4.8060)^2 \approx 23.10 = F$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$23.1 > 4.75 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim that the population means are equal.

🎯 Result

(a) There is sufficient evidence to reject the claim that the population means are equal.

(b) $t = -12.8285$

(c) There is sufficient evidence to reject the claim that the population means are equal.

PROBLEM STATEMENT

Chapter 12, Page 446

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

The P-value of the one-way ANOVA test is given in the row "Between groups" and in the column "p-value" of the ANOVa table in the exercise prompt:

$$P = 0.001632$$

If the P-value is smaller than the significance level, then we reject the null hypothesis:

$$P = 0.001632 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean waiting times at the four body shops.

Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 446

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

The P-value of the one-way ANOVA test is given in the row "Between groups" and in the column "p-value" of the ANOVa table in the exercise prompt:

$$P = 0.001866$$

If the P-value is smaller than the significance level, then we reject the null hypothesis:

$$P = 0.001866 < 0.01 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean mileage of the three cars.

Result

There is sufficient evidence to support the claim of a difference in the mean mileage of the three cars.

PROBLEM STATEMENT

Chapter 12, Page 447

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$c = \text{Confidence level} = 99\% = 0.99$$

(a) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

The P-value of the one-way ANOVA test is given in the row "Between groups" and in the column "p-value" of the ANOVA table in the exercise prompt:

$$P = 0.001005$$

If the P-value is smaller than the significance level, then we reject the null hypothesis:

$$P = 0.001005 < 0.01 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean production of the three assembly lines.

Step 2

(b) The degrees of freedom of error is given in the row "Withing groups" and in the column "df" in the ANOVA table in the exercise prompt:

$$d.f._E = 15$$

The mean square error is given in the row "Withing groups" and in the column "MS" in the ANOVA table in the exercise prompt:

$$MSE = 0.544444$$

The sample sizes are given in the column "Count" of the summary table in the exercise prompt:

$$n_1 = 6$$
$$n_2 = 6$$
$$n_3 = 6$$

The sample means are given in the column "Average" of the summary table in the exercise prompt:

$$\overline{x}_1 = 41.66667$$
$$\overline{x}_2 = 43.33333$$
$$\overline{x}_3 = 41.5$$

Step 3

Determine the critical value using the Student's T distribution table in the appendix, which is given in the row $df = 15$ and in the column $\alpha/2 = (1-c)/2 = 0.005$:

$$t_{\alpha/2} = 2.947$$

Determine the margin of error:

$$\begin{aligned} E &= t_{\alpha/2}\sqrt{MSE\left(\frac{1}{n_1}+\frac{1}{n_2}\right)} \\ &= 2.947\sqrt{0.544444\left(\frac{1}{6}+\frac{1}{6}\right)} \\ &\approx 1.25544 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} (\overline{x}_1 - \overline{x}_2) - E &= (43.33333 - 41.5) - 1.25544 \\ &= 1.83333 - 1.25544 \\ &= 0.57789 \end{aligned} \qquad \begin{aligned} (\overline{x}_1 - \overline{x}_2) + E &= (43.33333 - 41.5) + 1.25544 \\ &= 1.83333 + 1.25544 \\ &= 3.08877 \end{aligned}$$

Thus a 99% confidence interval for the difference in the means between Line B and Line C is $(0.57789, 3.08877)$.

Result

(a) There is sufficient evidence to support the claim of a difference in the mean production of the three assembly lines.

(b) $(0.57789, 3.08877)$

PROBLEM STATEMENT

Chapter 12, Page 447

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

The P-value of the one-way ANOVA test is given in the row "Error" and in the column "p-value" of the ANOVA table in the exercise prompt:

$$P = 0.051$$

If the P-value is smaller than the significance level, then we reject the null hypothesis:

$$P = 0.051 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the means for the different types of first-class mail.

Result

There is not sufficient evidence to support the claim of a difference in the means for the different types of first-class mail.

PROBLEM STATEMENT

Chapter 12, Page 448

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6 = \mu_7$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7 \text{ are equal}$$

The P-value of the one-way ANOVA test is given in the row "Error" and in the column "p-value" of the ANOVA table in the exercise prompt:

$$P = 0.000$$

If the P-value is smaller than the significance level, then we reject the null hypothesis:

$$P = 0.000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the means for the different days for the different days of the week.

Result

There is sufficient evidence to support the claim of a difference in the means for the different days for the different days of the week.

PROBLEM STATEMENT

Chapter 12, Page 448

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 4$$
$$b = \text{Number of blocks} = 3$$
$$n = \text{Total sample size} = 12$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (4-1)(3-1) = (3)(2) = 6$$

The F- critical value is given in the row with $dfd = d.f._E = 6$ and in the column with $dfn = d.f._T = 3$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.76$$

We reject the null hypothesis H_0 for treatments when the value of the test statistic exceeds 4.76.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (4-1)(3-1) = (3)(2) = 6$$

The F- critical value is given in the row with $dfd = d.f._E = 6$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 5.14$$

We reject the null hypothesis H_0 for blocks when the value of the test statistic exceeds 5.14.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 418 - \frac{66^2}{4 \cdot 3} \approx 55$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{11^2}{3} + \frac{15^2}{3} + \frac{17^2}{3} + \frac{23^2}{3} - \frac{66^2}{4 \cdot 3} \\ &\approx 25 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{16^2}{4} + \frac{21^2}{4} + \frac{29^2}{4} - \frac{66^2}{4 \cdot 3} \\ &\approx 21.5 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 55 - 25 - 21.5 = 8.5$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{25}{3} = 8.3333$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{21.5}{2} = 10.75$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{8.5}{6} = 1.4167$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{8.3333}{1.4167} \approx 5.88$$
$$F_B = \frac{MSB}{MSE} = \frac{10.75}{1.4167} \approx 7.59$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	25	3	8.3333	5.88
Blocks	21.5	2	10.75	7.59
Error	8.5	6	1.4167	
Total	55	11		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 5.88 > 4.76 \Rightarrow \text{ Reject } H_0$$
$$\textbf{Block: } 7.59 > 5.71 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the effectiveness of an advertisement by color.

There is sufficient evidence to support the claim of a difference in the effectiveness of an advertisement by size.

🎯 Result

There is sufficient evidence to support the claim of a difference in the effectiveness of an advertisement by color.

There is sufficient evidence to support the claim of a difference in the effectiveness of an advertisement by size.

PROBLEM STATEMENT

Chapter 12, Page 448

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 4$$
$$b = \text{Number of blocks} = 6$$
$$n = \text{Total sample size} = 24$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (4-1)(6-1) = (3)(5) = 15$$

The F- critical value is given in the row with $dfd = d.f._E = 15$ and in the column with $dfn = d.f._T = 3$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.29$$

We reject the null hypothesis H_0 for treatments when the value of the test statistic exceeds 3.29.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 6 - 1 = 5$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (4-1)(6-1) = (3)(5) = 15$$

The F- critical value is given in the row with $dfd = d.f._E = 15$ and in the column with $dfn = d.f._T = 5$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.90$$

We reject the null hypothesis H_0 for blocks when the value of the test statistic exceeds 2.90.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 1546932 - \frac{5848^2}{4 \cdot 6} \approx 121969.3333$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{1443^2}{6} + \frac{1380^2}{6} + \frac{1810^2}{6} + \frac{1215^2}{6} - \frac{5848^2}{4 \cdot 6} \\ &\approx 31533 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{794^2}{4} + \frac{1024^2}{4} + \frac{1250^2}{4} \\ &+ \frac{830^2}{4} + \frac{905^2}{4} + \frac{1045^2}{4} - \frac{5848^2}{4 \cdot 6} \\ &\approx 35402.8333 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 121969.3333 - 31533 - 35402.8333 = 55033.5$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{31533}{3} = 10511$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{35402.8333}{5} = 7080.5667$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{55033.5}{15} = 3668.9$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{10511}{3668.9} \approx 2.86$$
$$F_B = \frac{MSB}{MSE} = \frac{7080.5667}{3668.9} \approx 1.93$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	31533	3	10511	2.86
Blocks	35402.8333	5	7080.5667	1.93
Error	55033.5	15	3668.9	
Total	121969.3333	23		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 2.86 < 3.29 \Rightarrow \text{ Fail to reject } H_0$$
$$\textbf{Block: } 1.93 < 2.90 \Rightarrow \text{ Fail to reject } H_0$$

(a) There is not sufficient evidence to support the claim of a difference in the treatment means.

(b) There is not sufficient evidence to support the claim of a difference in the block means.

Result

(a) No

(b) No

PROBLEM STATEMENT

Chapter 12, Page 449

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 4$$
$$b = \text{Number of blocks} = 5$$
$$n = \text{Total sample size} = 20$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (4-1)(5-1) = (3)(4) = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 3$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.49$$

We reject the null hypothesis H_0 for treatments when the value of the test statistic exceeds 3.49.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 5 - 1 = 4$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (4-1)(5-1) = (3)(4) = 12$$

The F- critical value is given in the row with $dfd = d.f._E 126$ and in the column with $dfn = d.f._T = 4$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.26$$

We reject the null hypothesis H_0 for blocks when the value of the test statistic exceeds 3.26.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 79582 - \frac{1226^2}{4 \cdot 5} \approx 4428.2$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{305^2}{5} + \frac{315^2}{5} + \frac{305^2}{5} + \frac{301^2}{5} - \frac{1226^2}{4 \cdot 5} \\ &\approx 21.4 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{202^2}{4} + \frac{206^2}{4} + \frac{200^2}{4} \\ &+ \frac{267^2}{4} + \frac{351^2}{4} - \frac{1226^2}{4 \cdot 5} \\ &\approx 4278.7 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 4428.2 - 21.4 - 4278.7 = 128.1$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{21.4}{3} = 7.1333$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{4278.7}{4} = 1069.675$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{128.1}{12} = 10.675$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{7.1333}{10.675} \approx 0.67$$
$$F_B = \frac{MSB}{MSE} = \frac{1069.675}{10.675} \approx 100.20$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	21.4	3	7.1333	0.67
Blocks	4278.4	4	1069.675	100.20
Error	128.1	12	10.675	
Total	4428.2	19		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 0.67 < 3.49 \Rightarrow \text{ Fail to reject } H_0$$
$$\textbf{Block: } 100.20 > 3.26 \Rightarrow \text{ Reject } H_0$$

(a) There is not sufficient evidence to support the claim of a difference in the treatment means.

(b) There is sufficient evidence to support the claim of a difference in the block means.

Result

(a) No

(b) Yes

PROBLEM STATEMENT

Chapter 12, Page 449

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$b = \text{Number of blocks} = 4$$
$$n = \text{Total sample size} = 12$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(4-1) = (2)(3) = 6$$

The F- critical value is given in the row with $dfd = d.f._E = 6$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 5.14$$

We reject the null hypothesis H_0 for treatments when the value of the test statistic exceeds 5.14.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 4 - 1 = 3$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(4-1) = (2)(3) = 6$$

The F- critical value is given in the row with $dfd = d.f._E = 6$ and in the column with $dfn = d.f._T = 3$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.76$$

We reject the null hypothesis H_0 for blocks when the value of the test statistic exceeds 4.76.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 4890.83 - \frac{241.7^2}{3 \cdot 4} \approx 22.5892$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{78.9^2}{4} + \frac{79^2}{4} + \frac{83.8^2}{4} - \frac{241.7^2}{3 \cdot 4} \\ &\approx 3.9217 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{64.7^2}{3} + \frac{57.1^2}{3} + \frac{60.6^2}{3} \\ &+ \frac{59.3^2}{3} - \frac{241.7^2}{3 \cdot 4} \\ &\approx 10.2092 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 22.5892 - 3.9217 - 10.2092 = 8.4583$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{3.9217}{2} = 1.9608$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{10.2092}{3} = 3.4031$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{8.4583}{6} = 1.4097$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{1.9608}{1.4097} \approx 1.39$$
$$F_B = \frac{MSB}{MSE} = \frac{3.4031}{1.4097} \approx 2.41$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	3.9217	2	1.9608	1.39
Blocks	10.2092	3	3.4031	2.41
Error	8.4583	6	1.4097	
Total	22.5892	11		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 1.39 < 5.14 \Rightarrow \text{ Fail to reject } H_0$$
$$\textbf{Block: } 2.41 < 4.76 \Rightarrow \text{ Fail to reject } H_0$$

(a) There is not sufficient evidence to support the claim of a different among types of gasoline.

(b) There is not sufficient evidence to support the claim of a difference in the cars.

Result

(a) No

(b) No

PROBLEM STATEMENT

Chapter 12, Page 449

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$b = \text{Number of blocks} = 7$$
$$n = \text{Total sample size} = 21$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(7-1) = (2)(6) = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.89$$

We reject the null hypothesis H_0 for treatments when the value of the test statistic exceeds 3.89.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6 = \mu_7$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 7 - 1 = 6$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(7-1) = (2)(6) = 12$$

The F- critical value is given in the row with $dfd = d.f._E = 12$ and in the column with $dfn = d.f._T = 6$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.00$$

We reject the null hypothesis H_0 for blocks when the value of the test statistic exceeds 3.00.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 13812 - \frac{536^2}{3 \cdot 7} \approx 131.2381$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{146^2}{7} + \frac{144^2}{7} + \frac{161^2}{7} - \frac{536^2}{3 \cdot 7} \\ &\approx 44.0952 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{70^2}{3} + \frac{70^2}{3} + \frac{76^2}{3} + \frac{74^2}{3} \\ &+ \frac{82^2}{3} + \frac{82^2}{3} + \frac{77^2}{3} - \frac{536^2}{3 \cdot 7} \\ &\approx 77.2381 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 131.2381 - 44.0952 - 77.2381 = 9.9048$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{44.0952}{2} = 22.0476$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{77.2381}{6} = 12.8730$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{9.9048}{12} = 0.8254$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{22.0476}{0.8254} \approx 26.71$$
$$F_B = \frac{MSB}{MSE} = \frac{12.8730}{0.8254} \approx 15.60$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	44.0952	2	22.0476	26.71
Blocks	77.2381	6	12.8730	15.60
Error	9.9048	12	0.8254	
Total	131.2381	20		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 26.71 > 3.89 \Rightarrow \text{ Reject } H_0$$
$$\textbf{Block: } 15.60 > 3.00 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the gasolines. There is sufficient evidence to support the claim of a difference in the automobiles.

🎯 Result

There is sufficient evidence to support the claim of a difference in the gasolines. There is sufficient evidence to support the claim of a difference in the automobiles.

PROBLEM STATEMENT

Chapter 12, Page 449

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 3$$
$$b = \text{Number of blocks} = 9$$
$$n = \text{Total sample size} = 27$$
$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all treatments means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b-1)(k-1) = (3-1)(9-1) = (2)(8) = 16$$

The F- critical value is given in the row with $dfd = d.f._E = 16$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.63$$

We reject the null hypothesis H_0 for treatments when the value of the test statistic exceeds 3.63.

Step 2

The null hypothesis states that all block means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = ... = \mu_9$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, ..., \mu_9 \text{ are equal}$$

$d.f._B$ is the number of blocks b decreased by 1.

$$d.f._B = b - 1 = 9 - 1 = 8$$

$d.f._E$ is the product of the number of blocks and the number of groups, both decreased by 1.

$$d.f._E = (b - 1)(k - 1) = (3 - 1)(9 - 1) = (2)(8) = 16$$

The F- critical value is given in the row with $dfd = d.f._E = 16$ and in the column with $dfn = d.f._T = 8$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.59$$

We reject the null hypothesis H_0 for blocks when the value of the test statistic exceeds 2.59.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x^2 - \frac{(\sum x)^2}{bk} = 331.8218 - \frac{83.74^2}{3 \cdot 9} \approx 72.1037$$

Determine the value of the sum of squares between groups, where T_i represents the sum of all data values in treatment i:

$$\begin{aligned} SST &= \sum \frac{T_i^2}{b} - \frac{(\sum x)^2}{bk} \\ &= \frac{27.89^2}{9} + \frac{27.14^2}{9} + \frac{28.71^2}{9} - \frac{83.74^2}{3 \cdot 9} \\ &\approx 0.1370 \end{aligned}$$

Determine the value of the sum of squares within the blocks, where B_j represents the sum of all data values in block j:

$$\begin{aligned} SSB &= \sum \frac{B_j^2}{k} - \frac{(\sum x)^2}{bk} \\ &= \frac{3.21^2}{3} + \frac{3.45^2}{3} + \frac{5.77^2}{3} \\ &+ \frac{6.63^2}{3} + \frac{6.8^2}{3} + \frac{12.51^2}{3} \\ &+ \frac{15.05^2}{3} + \frac{13.48^2}{3} + \frac{16.84^2}{3} - \frac{83.74^2}{3 \cdot 9} \\ &\approx 71.6136 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups and decreased by the sum of squares within the blocks (Note: using this calculation you then immediately obtain that the sum $\text{Total } SS = SST + SSB + SSE$ holds):

$$SSE = \text{Total } SS - SST - SSB = 72.1037 - 0.1370 - 71.6136 = 0.3531$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{0.1370}{2} = 0.0685$$

MSB is SSB divided by $d.f._B$:

$$MSB = \frac{SSB}{d.f._B} = \frac{71.6136}{8} = 8.9517$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{0.3531}{16} = 0.0221$$

The value of the test statistic F_T is then MST divided by MSE, while the value F_B is MSB divided by MSE:

$$F_T = \frac{MST}{MSE} = \frac{0.0685}{0.0221} \approx 3.10$$
$$F_B = \frac{MSB}{MSE} = \frac{8.9517}{0.0221} \approx 405.62$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	0.1370	2	0.0685	3.10
Blocks	71.6136	8	8.9517	405.62
Error	0.3531	16	0.0221	
Total	72.1037	26		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$\textbf{Treatment: } 3.10 < 3.63 \Rightarrow \text{ Fail to reject } H_0$$
$$\textbf{Block: } 405.62 > 2.59 \Rightarrow \text{ Reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean prices at the supermarkets.

There is sufficient evidence to support the claim of a difference in the mean prices for the items.

🎯 Result

There is not sufficient evidence to support the claim of a difference in the mean prices at the supermarkets.

There is sufficient evidence to support the claim of a difference in the mean prices for the items.

PROBLEM STATEMENT

Chapter 12, Page 450

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 6 \\
n_1 &= \text{Sample size} = 12 \\
n_2 &= \text{Sample size} = 10 \\
n_3 &= \text{Sample size} = 14 \\
n_4 &= \text{Sample size} = 10 \\
n_5 &= \text{Sample size} = 8 \\
n_6 &= \text{Sample size} = 10 \\
n &= n_1 + n_2 + n_3 + n_4 + n_5 + n_6 = 12 + 10 + 14 + 10 + 8 + 10 = 64 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6 \text{ are equal}$$

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 6 - 1 = 5$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 64 - 6 = 58$$

The F- critical value is given in the row with $dfd = d.f._E = 58$ and in the column with $dfn = d.f._T = 5$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.37$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.37.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 53.4456 - \frac{58.4090^2}{64} \approx 0.1392$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{11.414^2}{6} + \frac{9.108^2}{6} + \frac{12.953^2}{6} \\ &+ \frac{8.831^2}{6} + \frac{7.182^2}{6} + \frac{8.921^2}{6} - \frac{58.4090^2}{64} \\ &\approx 0.0348 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 0.1392 - 0.0348 = 0.1044$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{0.0348}{5} = 0.0070$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{0.1044}{58} = 0.0018$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{0.0070}{0.0018} \approx 3.86$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	0.0348	5	0.0070	3.86
Error	0.1044	58	0.0018	
Total	0.1392	63		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$3.86 > 2.37 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean weights of candies of different colors.

🎯 Result

There is sufficient evidence to support the claim of a difference in the mean weights of candies of different colors.

PROBLEM STATEMENT

Chapter 12, Page 450

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 4 \\
n_1 &= \text{Sample size} = 10 \\
n_2 &= \text{Sample size} = 10 \\
n_3 &= \text{Sample size} = 10 \\
n_4 &= \text{Sample size} = 10 \\
n &= n_1 + n_2 + n_3 + n_4 = 10 + 10 + 10 + 10 = 40 \\
\alpha &= \text{Significance level} = 0.05 \\
c &= \text{Confidence level} = 95\% = 0.95 \\
\overline{x}_1 &= 51.32 \\
\overline{x}_2 &= 44.64 \\
\overline{x}_3 &= 47.2 \\
\overline{x}_4 &= 50.85 \\
SS \text{ total} &= 650.75
\end{aligned}
$$

(a) The overall mean is the sum of all values divided by the number of values:

$$
\begin{aligned}
\overline{x} &= \frac{n_1\overline{x}_1 + n_2\overline{x}_2 + n_3\overline{x}_3 + n_4\overline{x}_4}{n} \\
&= \frac{10(51.32) + 10(44.64) + 10(47.2) + 10(50.85)}{40} \\
&\approx 48.5025
\end{aligned}
$$

Determine the value of the sum of squares between groups:

$$
\begin{aligned}
SST &= \sum_{all\ groups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\
&= \sum_{all\ groups} n_i\overline{x}_i^2 - n\overline{x}^2 \\
&= 10(51.32)^2 + 10(44.64)^2 + 10(47.2)^2 + 10(50.85)^2 - 40(48.5025)^2 \\
&\approx 300.64475
\end{aligned}
$$

Step 2

(b) The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 650.75 - 300.64475 = 350.10525$$

Step 3

(c) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 40 - 4 = 36$$

The F- critical value is given in the row with $dfd = d.f._E = 36$ and in the column with $dfn = d.f._T = 3$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.87$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.87.

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{300.64475}{3} = 100.2149$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{350.10525}{36} = 9.7251$$

The value of the test statistic F is then MST divided by MSE:

$$F = \frac{MST}{MSE} = \frac{100.2149}{9.7251} \approx 10.30$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	300.64475	3	100.2149	10.30
Error	350.10525	36	9.7251	
Total	650.75	39		

Step 5

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$10.30 > 2.87 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the treatment means.

Step 6

(e) Determine the critical value using the Student's T distribution table in the appendix, which is given in the row $df = n - k = 40 - 4 = 36$ and in the column $\alpha/2 = (1 - c)/2 = 0.025$:

$$t_{\alpha/2} = 2.028$$

Determine the margin of error:

$$\begin{aligned} E &= t_{\alpha/2}\sqrt{MSE\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ &= 2.028\sqrt{9.7251\left(\frac{1}{10} + \frac{1}{10}\right)} \\ &\approx 2.8283 \end{aligned}$$

The boundaries of the confidence interval then become:

$$(\overline{x}_1 - \overline{x}_4) - E = (51.32 - 50.85) - 2.8283 = 0.47 - 2.8283 = -2.3583$$
$$(\overline{x}_1 - \overline{x}_4) + E = (51.32 - 50.85) + 2.8283 = 0.47 + 2.8283 = 3.2983$$

Since the confidence interval $(-2.3583, 3.2983)$ contains 0, there is not sufficient evidence to support the claim of a difference in the mean amount of music time between station 1 and station 4.

🎯 Result

(a) 300.64475

(b) 350.10525

(c)

Source	SS	df	MS	F
Treatments	300.64475	3	100.2149	10.30
Error	350.10525	36	9.7251	
Total	650.75	39		

(d) Yes

(e) No

Exercise 47

PROBLEM STATEMENT

Chapter 12, Page 450

Solution A

Step 1

Given:

$$k = \text{Number of groups} = 2$$
$$b = \text{Number of blocks} = 2$$
$$r = \text{Number of replications} = 5$$
$$n = \text{Total sample size} = 2 \times 2 \times 5 = 20$$
$$\alpha = \text{Significance level} = 0.05$$

(a) Let us first determine the mean of all six combinations of machine and position. The mean is the sum of all data values divided by the number of data values.

Step 2

$$\overline{x}_{11} = \frac{\sum x_i}{n} = \frac{978 + 1035 + 964 + 996 + 1117}{5} = 1018$$
$$\overline{x}_{12} = \frac{\sum x_i}{n} = \frac{1335 + 1167 + 1236 + 1317 + 1192}{5} = 1249.4$$
$$\overline{x}_{21} = \frac{\sum x_i}{n} = \frac{863 + 975 + 999 + 1019 + 1037}{5} = 978.6$$
$$\overline{x}_{22} = \frac{\sum x_i}{n} = \frac{1079 + 1160 + 1063 + 1110 + 1093}{5} = 1101$$

(b) I used technology to determine the following two-way ANOVA table (for example, the website vassarstats.net contains a free two-way ANOVA calculator that will calculate the ANOVA table below)

Source	df	SS	MS	F	P
Rows	1	44086.05	44086.05	11.44	0.0038
Columns	1	156468.05	40.61	<0.0001	
r x c	1	14851.25	14851.25	3.85	0.0674
Error	16	61640.4	3852.53		
Total	19	277045.75			

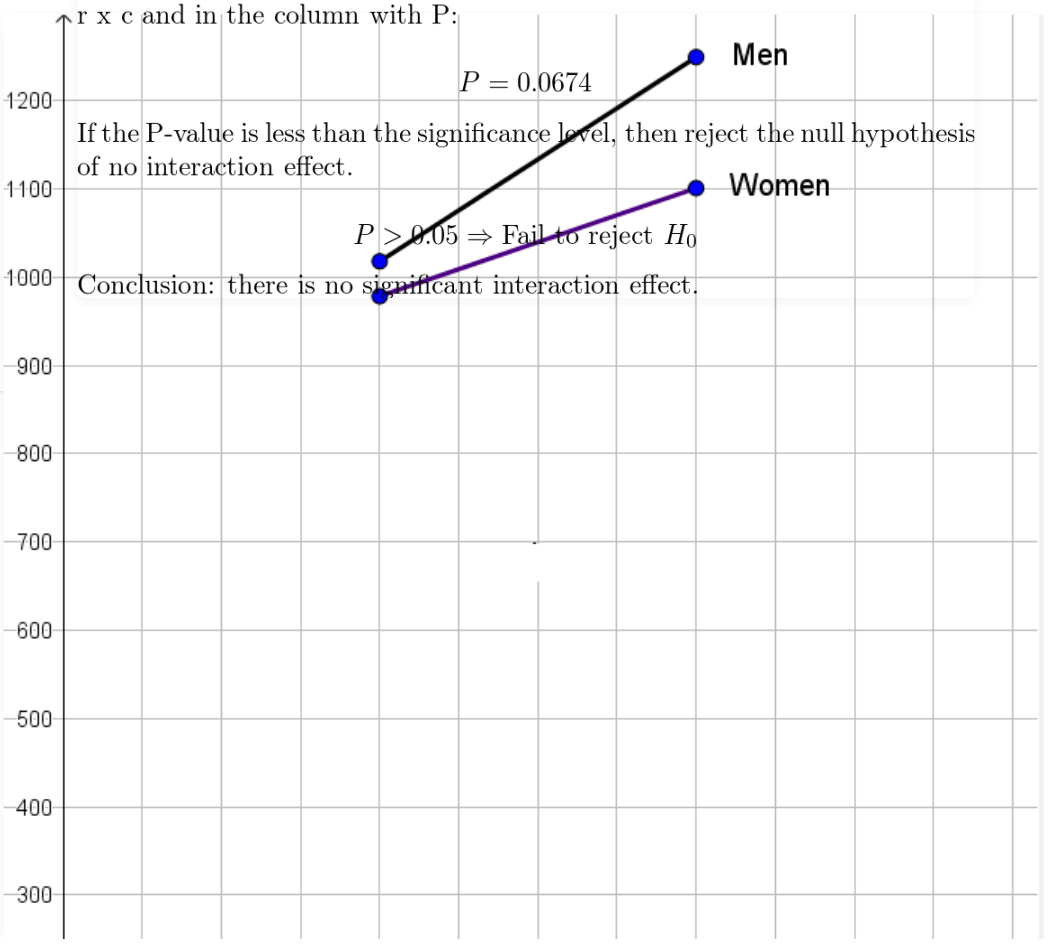
Next, we create the interaction graph. The machine is on the horizontal axis, while the mean is on the vertical axis. We connect consecutive points corresponding to the same positions with a straight line.

We note that the lines are approximately parallel in the interaction graph below, which indicates that there is no interaction effect.

Interaction effect

The mean wage of men appears to be higher than the mean wage of women.

The P-value of the interaction effect is given in the output in the row with r x c and in the column with P:



If the P-value is less than the significance level, then reject the null hypothesis of no interaction effect.

Conclusion: there is no significant interaction effect.



Step 3

Gender

The P-value for the row variable is given in the output in the row with ”Rows” and in the column with P:

$$P = 0.0038$$

If the P-value is less than the significance level, then reject the null hypothesis.

$$0.0038 < 0.05 \Rightarrow \text{Reject } H_0$$

Conclusion: There is a significant effect from the row factor "Gender".

Sector

The P-value for the row variable is given in the output in the row with ”Columns” and in the column with P:

$$P < 0.0001$$

If the P-value is less than the significance level, then reject the null hypothesis.

$$P < 0.05 \Rightarrow \text{ Reject } H_0$$

Conclusion: There is a significant effect from the column factor "Sector".

Step 4

(c) We need to compare the mean wage for men and women within each sector, which is done by constructing a one-way ANOVA table per sector (for example, the website vassarstats.net contains a free one-way ANOVA calculator that will calculate the ANOVA tables below).

Public

Source	df	SS	MS	F	P
Gender	1	3880.9	3880.9	0.92	0.365562
Error	8	33925.2	4240.65		
Total	9	37806.1			

Private

Source	df	SS	MS	F	P
Gender	1	55056.4	55056.4	15.89	0.004026
Error	8	27715.2	3464.4		
Total	9	82771.6			

Step 5

If the P-value is less than the significance level, then reject the null hypothesis of an effect of Position on the mean sales.

$$0.366562 > 0.05 \Rightarrow \text{Fail to reject } H_0$$
$$0.004026 < 0.05 \Rightarrow \text{Reject } H_0$$

Conclusion: There is no significant effect of Gender on the public sector, while there is an effect of Gender on the private sector.

Step 6

(d) We then recommend hiring females in the private sector (as they have lower mean wages), while the gender doesn't matter in the public sector.

🎯 Result

(a) No

(b) No significant interaction effect, Significant effect of Gender, Significant effect of Sector

(c) There is no significant effect of Gender on the public sector, while there is an effect of Gender on the private sector.

(d) We recommend hiring females in the private sector (as they have lower mean wages), while the gender doesn't matter in the public sector.

PROBLEM STATEMENT

Chapter 12, Page 451

Solution A

Step 1

Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 4 \\
b &= \text{Number of blocks} = 4 \\
r &= \text{Number of replications} = 2 \\
n &= \text{Total sample size} = 4 \times 4 \times 2 = 32 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

(a) Let us first determine the mean of all six combinations of machine and position. The mean is the sum of all data values divided by the number of data values.

$$
\begin{aligned}
\overline{x}_{11} &= \frac{\sum x_i}{n} = \frac{0.13 + 0.11}{2} = 0.12 \\
\overline{x}_{12} &= \frac{\sum x_i}{n} = \frac{0.12 + 0.11}{2} = 0.115 \\
\overline{x}_{13} &= \frac{\sum x_i}{n} = \frac{0.19 + 0.17}{2} = 0.18 \\
\overline{x}_{14} &= \frac{\sum x_i}{n} = \frac{0.15 + 0.18}{2} = 0.165 \\
\overline{x}_{21} &= \frac{\sum x_i}{n} = \frac{0.14 + 0.10}{2} = 0.12 \\
\overline{x}_{22} &= \frac{\sum x_i}{n} = \frac{0.15 + 0.14}{2} = 0.145 \\
\overline{x}_{23} &= \frac{\sum x_i}{n} = \frac{0.18 + 0.17}{2} = 0.175 \\
\overline{x}_{24} &= \frac{\sum x_i}{n} = \frac{0.20 + 0.18}{2} = 0.19 \\
\overline{x}_{31} &= \frac{\sum x_i}{n} = \frac{0.16 + 0.17}{2} = 0.165 \\
\overline{x}_{32} &= \frac{\sum x_i}{n} = \frac{0.15 + 0.14}{2} = 0.145 \\
\overline{x}_{33} &= \frac{\sum x_i}{n} = \frac{0.18 + 0.19}{2} = 0.185 \\
\overline{x}_{34} &= \frac{\sum x_i}{n} = \frac{0.19 + 0.21}{2} = 0.2
\end{aligned}
$$

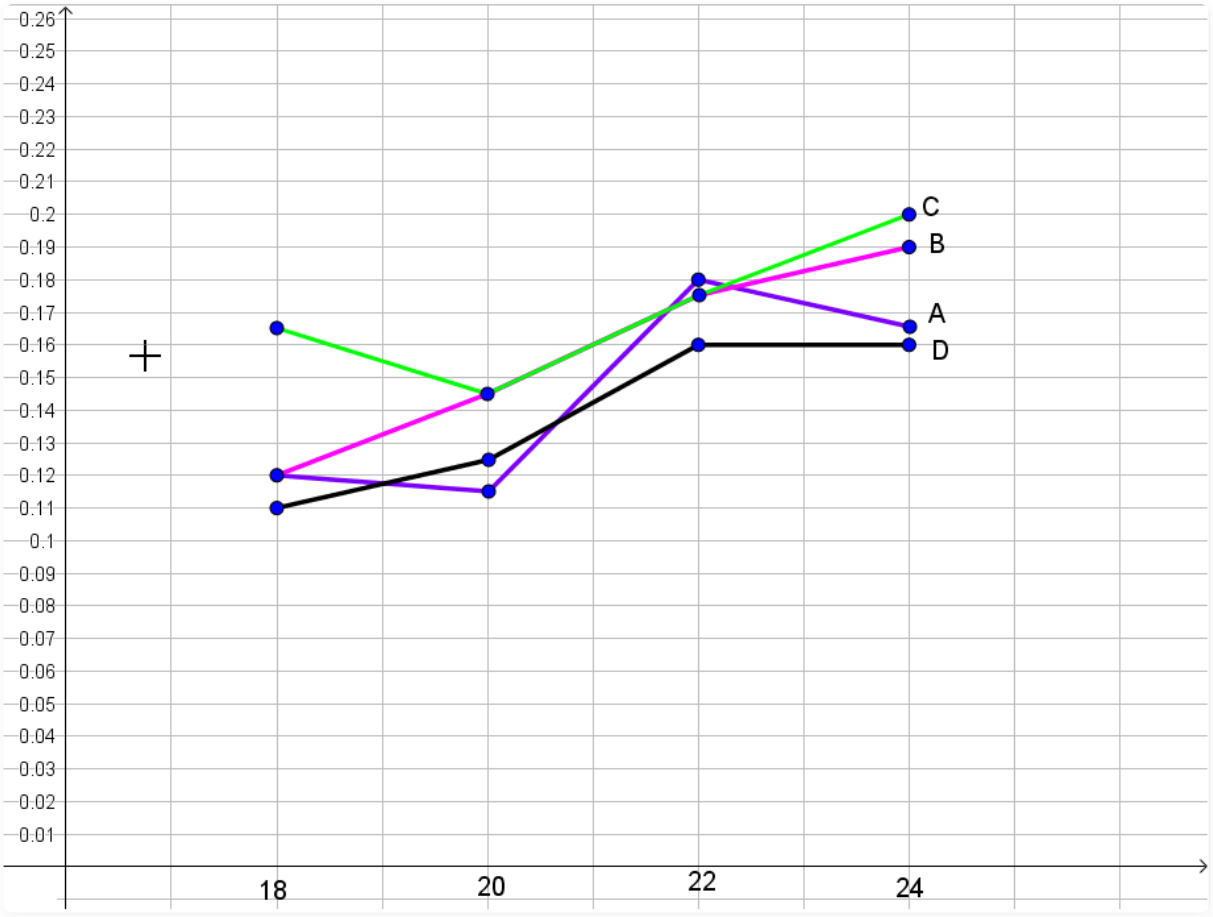
Step 2

$$\overline{x}_{41} = \frac{\sum x_i}{n} = \frac{0.09 + 0.13}{2} = 0.11$$
$$\overline{x}_{42} = \frac{\sum x_i}{n} = \frac{0.12 + 0.13}{2} = 0.125$$
$$\overline{x}_{43} = \frac{\sum x_i}{n} = \frac{0.16 + 0.16}{2} = 0.16$$
$$\overline{x}_{44} = \frac{\sum x_i}{n} = \frac{0.15 + 0.17}{2} = 0.16$$

Next, we create the interaction graph. The machine is on the horizontal axis, while the mean is on the vertical axis. We connect consecutive points corresponding to the same positions with a straight line.

We note that the lines intersect in the interaction graph below, which indicates that there is an interaction effect.

Moreover, the mean pounds of dirt removed or detergent A appear to be highest at a cycle time of 22 minutes and lowest at a cycle time of 20 minutes, the mean pounds of dirt removed or detergent B appear to be highest at a cycle time of 24 minutes and lowest at a cycle time of 18 minutes, the mean pounds of dirt removed or detergent C appear to be highest at a cycle time of 24 minutes and lowest at a cycle time of 20 minutes, and the mean pounds of dirt removed or detergent D appear to be highest at a cycle time of 22/24 minutes and lowest at a cycle time of 18 minutes.



Step 3

(b) I used technology to determine the following two-way ANOVA table (for example, the website vassarstats.net contains a free two-way ANOVA calculator that will calculate the ANOVA table below).

Source	df	SS	MS	F	P
Rows	3	0.017225	0.005742	27.019608	<0.0001
Columns	3	0.005725	0.001908	8.890392	0.0010
r x c	9	0.002200	0.000224	0.385968	
Error	16	0.00485	0.000303		
Total	31	0.03			

Interaction effect

The P-value of the interaction effect is given in the output in the row with r x c and in the column with P:

$$P = 0.385968$$

If the P-value is less than the significance level, then reject the null hypothesis of no interaction effect.

$$P > 0.05 \Rightarrow \text{Fail to reject } H_0$$

Conclusion: there is no interaction effect.

Step 4

Detergent brand

The P-value for the row variable is given in the output in the row with ”Rows” and in the column with P:

$$P = 0.0010$$

If the P-value is less than the significance level, then reject the null hypothesis.

$$0.0010 < 0.05 \Rightarrow \text{Reject } H_0$$

Conclusion: There is a significant effect from the row factor "Detergent brand".

Machine

The P-value for the row variable is given in the output in the row with ”Columns” and in the column with P:

$$P < 0.0001$$

If the P-value is less than the significance level, then reject the null hypothesis.

$$P < 0.05 \Rightarrow \text{ Reject } H_0$$

Conclusion: There is a significant effect from the column factor "Machine".

Step 5

(c) We need to compare the mean pounds of dirt removed for each brand, which is done by constructing a one-way ANOVA table per machine (for example, the website vassarstats.net contains a free one-way ANOVA calculator that will calculate the ANOVA tables below).

Brand A

Source	df	SS	MS	F	P
Cycle time	3	0.0063	0.0021	9.33	0.028063
Error	4	0.0009	0.000225		
Total	7	0.0072			

Brand B

Source	df	SS	MS	F	P
Cycle time	3	0.000585	0.00195	7.09	0.044422
Error	4	0.0011	0.000275		
Total	7	0.00695			

Brand C

Source	df	SS	MS	F	P
Cycle time	3	0.003437	0.001146	13.1	0.015499
Error	4	0.00035	0.000087		
Total	7	0.003787			

Step 6

Brand D

Source	df	SS	MS	F	P
Cycle time	3	0.003837	0.001279	4.87	0.080142
Error	4	0.00105	0.000263		
Total	7	0.004887			

If the P-value is less than the significance level, then reject the null hypothesis of an effect of Position on the mean sales.

$$0.028063 < 0.05 \Rightarrow \text{Reject } H_0$$
$$0.044422 < 0.05 \Rightarrow \text{Reject } H_0$$
$$0.015499 < 0.05 \Rightarrow \text{Reject } H_0$$
$$0.080142 > 0.05 \Rightarrow \text{Fail to reject } H_0$$

Conclusion: There is no significant effect of Cycle time on brand D, while there is an effect of cycle time on the other three brands.

Step 7

(d) We then recommend using a cycle time of 22 minutes for brand A, a cycle time of 24 minutes for brand B, a cycle time of 24 minutes for brand D, while the cycle time doesn't matter for brand D.

🎯 Result

(a) Yes

(b) No significant interaction effect, Significant effect of both factors

(c) There is no significant effect of Cycle time on brand D, while there is an effect of cycle time on the other three brands.

(d) We recommend using a cycle time of 22 minutes for brand A, a cycle time of 24 minutes for brand B, a cycle time of 24 minutes for brand D, while the cycle time doesn't matter for brand D.

PROBLEM STATEMENT

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Solution A

Step 1

(a) Given:

$$n_2 = \text{Sample size} = 38$$
$$n_1 = \text{Sample size} = 67$$
$$\alpha = \text{Significance level} = 0.02$$

The mean is the sum of all values divided by the number of values:

$$\overline{x}_2 = \frac{224 + 206 + 179 + ... + 175.6}{38} = 202.7974$$
$$\overline{x}_1 = \frac{243.7 + 246 + 270.8 + ... + 139.9}{67} = 231.4851$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_2 = \sqrt{\frac{(224 - 202.7974)^2 + ... + (175.6 - 202.7974)^2}{38 - 1}} \approx 33.7051$$
$$s_1 = \sqrt{\frac{(243.7 - 231.4851)^2 + ... + (139.9 - 231.4851)^2}{67 - 1}} \approx 50.5693$$

Step 2

Given claim: Variation differs

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{50.5693^2}{33.7051^2} \approx 2.2510$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 38 - 1 = 37$ and in the column with $dfn = n_1 - 1 = 67 - 1 = 66$ of the F distribution-table $\alpha/2 = 0.01$ in the appendix :

$$F_{\alpha/2} = 2.047$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 2.2510 > 2.047 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the variability of the selling prices of the homes that have a pool versus those that do not have a pool.

Step 3

(b) Given:

$$\begin{aligned} n_2 &= \text{Sample size} = 34 \\ n_1 &= \text{Sample size} = 71 \\ \alpha &= \text{Significance level} = 0.02 \end{aligned}$$

The mean is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_2 &= \frac{206 + 207.5 + 147.4 + ... + 139.9}{38} = 185.45 \\ \overline{x}_1 &= \frac{224 + 179 + 198.9 + ... + 189.4}{67} = 231.4851 \end{aligned}$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$\begin{aligned} s_2 &= \sqrt{\frac{(206 - 185.45)^2 + ... + (139.9 - 185.45)^2}{38 - 1}} \approx 28.0046 \\ s_1 &= \sqrt{\frac{(224 - 238.1761)^2 + ... + (189.4 - 238.1761)^2}{67 - 1}} \approx 44.8765 \end{aligned}$$

Step 4

Given claim: Variation differs

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{44.8765^2}{28.0046^2} \approx 2.5679$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 34 - 1 = 33$ and in the column with $dfn = n_1 - 1 = 71 - 1 = 70$ of the F distribution-table $\alpha/2 = 0.01$ in the appendix :

$$F_{\alpha/2} = 1.946$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 2.5679 > 1.946 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the variability of the selling prices of the homes with an attached garage versus those that do not have an attached garage.

Step 5

(c) Given:

$$
\begin{aligned}
k &= \text{Number of groups} = 5 \\
n_1 &= \text{Sample size} = 15 \\
n_2 &= \text{Sample size} = 20 \\
n_3 &= \text{Sample size} = 25 \\
n_4 &= \text{Sample size} = 29 \\
n_5 &= \text{Sample size} = 16 \\
n &= n_1 + n_2 + n_3 + n_4 + n_5 = 15 + 20 + 25 + 29 + 16 = 105 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3, \mu_4, \mu_5 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	15	20	25	29	16	105
ΣX	2953.7	4549	5719.8	6290.9	3702.4	23215.8
Mean	196.9133	227.45	228.792	216.9276	231.4	221.1029
ΣX^2	599549.89	1071778.08	1365459.06	1434605.75	892454.52	5363847.3
Variance	1280.4984	1953.0542	2367.2741	2497.7142	2381.2773	2218.9191
Std.Dev.	35.7841	44.1934	48.6546	49.9771	48.7983	47.1054
Std.Err.	9.2394	9.8819	9.7309	9.2805	12.1996	4.597

Step 6

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 5 - 1 = 4$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 105 - 5 = 100$$

The F- critical value is given in the row with $dfd = d.f._E = 100$ and in the column with $dfn = d.f._T = 4$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.46$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.46.

Step 7

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 5363847.3 - \frac{23215.8^2}{105} \approx 230767.5891$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{2953.7^2}{15} + \frac{4549^2}{20} + \frac{5719.8^2}{25} \\ &+ \frac{6290.9^2}{29} + \frac{3702.4^2}{16} - \frac{16800.4^2}{105} \\ &\approx 13262.8455 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 230767.5891 - 13262.8455 = 217504.7437$$

Step 8

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{13262.8455}{4} = 3315.7114$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{217504.7437}{100} = 2175.0474$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{3315.7114}{2175.0474} \approx 1.52$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	13262.8455	4	3315.7114	1.52
Error	217504.7437	100	2175.0474	
Total	230767.5891			

Step 9

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$1.52 < 2.46 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean selling price of the homes among the five townships.

🎯 Result

(a) Yes

(b) Yes

(c) No

PROBLEM STATEMENT

Chapter 12, Page 452

Solution A

Step 1

(a) Given:

$$n_2 = \text{Sample size} = 16$$
$$n_1 = \text{Sample size} = 14$$
$$\alpha = \text{Significance level} = 0.10$$

The mean is the sum of all values divided by the number of values:

$$\overline{x}_2 = \frac{36.8 + 43.7 + 48.7 + ... + 149.4}{16} = 84.3438$$
$$\overline{x}_1 = \frac{62.3 + 63.3 + 65.3 + ... + 201.5}{14} = 93.2786$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_2 = \sqrt{\frac{(36.8 - 84.3438)^2 + ... + (149.4 - 84.3438)^2}{16 - 1}} \approx 31.0924$$
$$s_1 = \sqrt{\frac{(62.3 - 93.2786)^2 + ... + (201.5 - 93.2786)^2}{14 - 1}} \approx 37.4496$$

Step 2

Given claim: Variation differs

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

Compute the value of the test statistic:

$$F = \frac{s_1^2}{s_2^2} = \frac{37.4496^2}{31.0924^2} \approx 1.4507$$

The F- critical value is given in the row with $dfd = n_2 - 1 = 16 - 1 = 15$ and in the column with $dfn = n_1 - 1 = 14 - 1 = 13$ of the F distribution-table $\alpha/2 = 0.05$ in the appendix :

$$F_{\alpha/2} = 2.448$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 1.4507 < 2.448 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the variation in the team salary among the American and National League teams.

Step 3

(b) Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 11 \\ n_2 &= \text{Sample size} = 10 \\ n_3 &= \text{Sample size} = 9 \\ n &= n_1 + n_2 + n_3 = 11 + 10 + 9 = 30 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	11	10	9			30
ΣX	789	834	807			2430
Mean	71.7273	83.4	89.6667			81
ΣX^2	57455	69980	73187			200622
Variance	86.2182	47.1556	103.25			130.7586
Std.Dev.	9.2854	6.867	10.1612			11.435
Std.Err.	2.7996	2.1715	3.3871			2.0877

Step 4

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 30 - 3 = 27$$

The F- critical value is given in the row with $dfd = d.f._E = 27$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.01$ in the appendix :

$$F_\alpha = 5.49$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 5.49.

Step 5

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 200622 - \frac{2430^2}{30} \approx 3792$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{789^2}{11} + \frac{454834.6^2}{10} + \frac{807^2}{9} - \frac{2430^2}{30} \\ &\approx 1679.4183 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 3792 - 1679.4183 = 2112.5818$$

Step 6

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{1679.4183}{2} = 839.7091$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{2112.5818}{27} = 78.2438$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{839.7091}{78.2438} \approx 10.73$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	1679.4182	2	839.7091	10.73
Error	2112.5818	27	78.2438	
Total	3792	29		

Step 7

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$10.73 > 5.49 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean number of games won among the three groups.

Step 8

(c) Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 11 \\ n_2 &= \text{Sample size} = 10 \\ n_3 &= \text{Sample size} = 9 \\ n &= n_1 + n_2 + n_3 = 11 + 10 + 9 = 30 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	11	10	9			30
ΣX	1747	1699	1596			5042
Mean	158.8182	169.9	177.3333			168.0667
ΣX^2	283851	295983	299280			879114
Variance	639.5636	813.6556	2032			1093.8575
Std.Dev.	25.2896	28.5246	45.0777			33.0735
Std.Err.	7.6251	9.0203	15.0259			6.0384

Step 9

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 30 - 3 = 27$$

The F- critical value is given in the row with $dfd = d.f._E = 27$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.01$ in the appendix :

$$F_\alpha = 5.49$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 5.49.

Step 10

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 879114 - \frac{5042^2}{30} \approx 31721.8667$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{1747^2}{11} + \frac{1669^2}{10} + \frac{1596^2}{9} - \frac{5042^2}{30} \\ &\approx 1747.3303 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 31721.8667 - 1747.3303 = 29974.5364$$

Step 11

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{1747.3303}{2} = 873.6652$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{29974.5364}{27} = 1110.168$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{873.6652}{1110.168} \approx 0.79$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	1747.3303	2	873.6652	0.79
Error	29974.5364	27	1110.168	
Total	31721.8667	29		

Step 12

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$0.79 < 5.49 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean number of home runs hit per team.

Step 13

(d) Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 11 \\ n_2 &= \text{Sample size} = 10 \\ n_3 &= \text{Sample size} = 9 \\ n &= n_1 + n_2 + n_3 = 11 + 10 + 9 = 30 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	11	10	9			30
ΣX	688.4	874.6	1092.4			2655.4
Mean	62.5818	87.46	121.3778			88.5133
ΣX^2	45188.32	79019.7	144159.54			268367.56
Variance	210.6996	280.7982	1445.8069			1149.2846
Std.Dev.	14.5155	16.757	38.0238			33.9011
Std.Err.	4.3766	5.299	12.6746			6.1895

Step 14

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 30 - 3 = 27$$

The F- critical value is given in the row with $dfd = d.f._E = 27$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.01$ in the appendix :

$$F_\alpha = 5.49$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 5.49.

Step 15

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 268367.56 - \frac{3655.4^2}{30} \approx 33329.2547$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{688.4^2}{11} + \frac{874.6^2}{10} + \frac{1092.4^2}{9} - \frac{3655.4^2}{30} \\ &\approx 17128.6187 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 33329.2547 - 17128.6187 = 16200.6359$$

Step 16

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{17128.6187}{2} = 8564.3094$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{16200.6359}{27} = 600.0236$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{8564.3094}{600.0236} \approx 14.27$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	17128.6187	2	8564.3094	14.27
Error	16200.6359	27	600.0236	
Total	33329.2547	29		

Step 17

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$14.27 > 5.49 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean salary of the three groups.

🎯 Result

(a) No

(b) Yes

(c) No

(d) Yes

PROBLEM STATEMENT

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Solution A

Step 1

(a) Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 47 \\ n_2 &= \text{Sample size} = 25 \\ n_3 &= \text{Sample size} = 8 \\ n &= n_1 + n_2 + n_3 = 47 + 25 + 8 = 80 \\ \alpha &= \text{Significance level} = 0.05 \\ c &= \text{Confidence level} = 95\% = 0.95 \end{aligned}$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Step 2

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 80 - 3 = 77$$

The F- critical value is given in the row with $dfd = d.f._E = 77$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.89$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.89.

Step 3

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 16448399 - \frac{36023^2}{80} \approx 227692.3875$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{20765^2}{47} + \frac{11204^2}{25} + \frac{4054^2}{8} - \frac{36023^2}{80} \\ &\approx 28996.2509 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 227692.3875 - 28996.2509 = 198696.1366$$

Step 4

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{28996.2509}{2} = 14498.1255$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{198696.1366}{77} = 2580.4693$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{14498.1255}{2580.4693} \approx 5.62$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	28996.2509	2	14498.1255	5.62
Error	198696.1366	77	2580.4693	
Total	227692.3875	79		

Step 5

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$5.62 > 4.89 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim that the mean maintenance cost is equal for each of the bus producers.

Step 6

(b) Given:

$$\begin{aligned} k &= \text{Number of groups} = 3 \\ n_1 &= \text{Sample size} = 47 \\ n_2 &= \text{Sample size} = 25 \\ n_3 &= \text{Sample size} = 8 \\ n &= n_1 + n_2 + n_3 = 47 + 25 + 8 = 80 \\ \alpha &= \text{Significance level} = 0.05 \\ c &= \text{Confidence level} = 95\% = 0.95 \end{aligned}$$

The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Data Summary

	Samples					
	1	2	3	4	5	Total
N	47	25	8			80
ΣX	39205	20469	6735			66409
Mean	834.1489	818.76	841.875			830.1125
ΣX^2	32804819	16790829	5671901			55267549
Variance	2217.6078	1317.94	267.5536			1779.8479
Std.Dev.	47.0915	36.3034	16.3571			42.1882
Std.Err.	6.869	7.2607	5.7831			4.7168

Step 7

$d.f._T$ is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 3 - 1 = 2$$

$d.f._E$ is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 80 - 3 = 77$$

The F- critical value is given in the row with $dfd = d.f._E = 77$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 4.89$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 4.89.

Step 8

Determine the value of total-group variability SS_{total}:

$$SS_{total} = \sum x_{total}^2 - \frac{(\sum x_{total})^2}{n} = 55267549 - \frac{66409^2}{80} \approx 140607.9875$$

Determine the value of the sum of squares between groups:

$$\begin{aligned} SST &= \sum_{allgroups} \frac{(\sum x_i)^2}{n_i} - \frac{(\sum x_{total})^2}{n} \\ &= \frac{39205^2}{47} + \frac{20469^2}{25} + \frac{6735^2}{8} - \frac{66409^2}{80} \\ &\approx 5094.5951 \end{aligned}$$

The value of the sum of squares within groups is then the value of the total-group variability decreased by the value of the sum of squares between groups (Note: using this calculation you then immediately obtain that the sum $SS_{total} = SST + SSE$ holds):

$$SSE = SS_{total} - SST = 140607.9875 - 5094.5951 = 135513.3924$$

Step 9

MST is SST divided by $d.f._T$:

$$MST = \frac{SST}{d.f._T} = \frac{5094.5951}{2} = 2547.2975$$

MSE is SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{135513.3924}{77} = 1759.9142$$

The value of the test statistic F is then MS_{Tr} divided by MS_E:

$$F = \frac{MST}{MSE} = \frac{2547.2975}{1759.9142} \approx 1.45$$

Combine the information in an ANOVA table:

Source	SS	df	MS	F
Treatments	5094.5951	2	2547.2975	1.45
Error	135513.3924	77	1759.9142	
Total	140607.9875	79		

Step 10

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$1.45 < 4.89 \Rightarrow \text{ Reject } H_0$$

There is not sufficient evidence to reject the claim that the mean miles traveled is equal for each make of bus.

Step 11

(c)

$$c = \text{Confidence interval} = 95\% = 0.95$$

Determine the critical value using the Student's T distribution table in the appendix, which is given in the row $df = n - k = 80 - 3 = 77$ and in the column $\alpha/2 = (1 - c)/2 = 0.025$:

$$t_{\alpha/2} = 1.991$$

Determine the margin of error:

$$\begin{aligned} E &= t_{\alpha/2}\sqrt{MSE\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ &= 1.991\sqrt{2580.4693\left(\frac{1}{47} + \frac{1}{8}\right)} \\ &\approx 38.6819 \end{aligned}$$

The boundaries of the confidence interval then become:

$$(\overline{x}_1 - \overline{x}_3) - E = (834.1489 - 841.875) - 38.6819 = -7.7261 - 38.6819 = -46.4080$$
$$(\overline{x}_1 - \overline{x}_3) + E = (834.1489 - 841.875) + 38.6819 = -7.7261 + 38.6819 = 30.9558$$

Result

(a) There is sufficient evidence to reject the claim that the mean maintenance cost is equal for each of the bus producers.

(b) There is not sufficient evidence to reject the claim that the mean miles traveled is equal for each make of bus.

(c) $(-46.4080, 30.9558)$

Statistics

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Chapter 12 **Exercise 1** Page 459

PROBLEM STATEMENT

Chapter 12, Page 459

Solution A

Step 1

The null hypothesis and the alternative hypothesis both state a claim about a population parameter.

However, the **null hypothesis** always needs to include an equality.

Result

Null hypothesis

Chapter 12 Exercise 2

Page 459

PROBLEM STATEMENT

Chapter 12, Page 459

Solution A

Step 1

The **significance level** is the probability of rejecting the null hypothesis $H_.$, when the null hypothesis H_0 is true.

Result

Significance level

Chapter 12 Exercise 3

Page 460

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

It is only appropriate to conduct a hypothesis test about a population proportion, when the number of successes in the sample is at least 5.

Since the sample proportion π is the number of successes divided by the sample size n, the value $n\pi$ represents the number of successes and thus the value $n\pi$ needs to be at least $\mathbf{5}$.

Result

5

Chapter 12 Exercise 4

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

When performing a hypothesis test about a single population mean, we need to use the z-distribution when the **population standard deviation** σ is known and we need to use the t-distribution when the population standard deviation σ is unknown.

Result

Population standard deviation

Chapter 12 Exercise 5

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

When conducting a two-sample hypothesis test about means with the population standard deviations unknown, we require that the population distributions are **approximately normally distributed** in order to use the t-distribution in the hypothesis test.

Result

Approximately normal

Chapter 12 Exercise 6

Page 460

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

The value of the **test statistic** is determined by using sample information and is used to determine whether to reject the null hypothesis H_0 or not.

Result

Test statistic

Chapter 12 Exercise 7

Page 460

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

In a two-tailed hypothesis test, we need to take into account both tails of the distributions and thus the rejection region will be **evenly split between the two tails**.

Result

Evenly split between the two tails

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

The F-distribution is a continuous probability distribution, which is positively skewed.

Moreover, the F-distribution represents a family of distribution, because different values for the degrees of freedom for the numerator and/or degrees of freedom for the denominator result in a different F-distribution.

However, the F-distribution does not range from $-\infty$ to $+\infty$, because the distribution only ranges from 0 to $+\infty$.

Result

Range from $-\infty$ to $+\infty$

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Two events are **independent**, if the probability that one event occurs in no way affects the probability of the other event occurring.

Two events are **disjoint** or **mutually exclusive**, if the events cannot occur at the same time.

When performing a one-way ANOVA test, we require that the treatments are *independent* such that one treatment won't influence other treatments.

Result

Independent

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of groups} = 4 \\ n_1 &= \text{Sample size} = 6 \\ n_2 &= \text{Sample size} = 6 \\ n_3 &= \text{Sample size} = 6 \\ n_4 &= \text{Sample size} = 6 \\ n &= n_1 + n_2 + n_3 + n_4 = 6 + 6 + 6 + 6 = 24 \end{aligned}$$

The degrees of freedom for the numerator is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

The degrees of freedom for the denominator is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 24 - 4 = 20$$

Result

3 and 20

Statistics

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Chapter 12 **Exercise 1** Page 459

PROBLEM STATEMENT

Chapter 12, Page 459

Solution A

Step 1

The null hypothesis and the alternative hypothesis both state a claim about a population parameter.

However, the **null hypothesis** always needs to include an equality.

Result

Null hypothesis

Chapter 12 Exercise 2

Page 459

PROBLEM STATEMENT

Chapter 12, Page 459

Solution A

Step 1

The **significance level** is the probability of rejecting the null hypothesis $H_.$, when the null hypothesis H_0 is true.

Result

Significance level

Chapter 12 Exercise 3

Page 460

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

It is only appropriate to conduct a hypothesis test about a population proportion, when the number of successes in the sample is at least 5.

Since the sample proportion π is the number of successes divided by the sample size n, the value $n\pi$ represents the number of successes and thus the value $n\pi$ needs to be at least $\mathbf{5}$.

Result

5

Chapter 12 Exercise 4

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

When performing a hypothesis test about a single population mean, we need to use the z-distribution when the **population standard deviation** σ is known and we need to use the t-distribution when the population standard deviation σ is unknown.

Result

Population standard deviation

Chapter 12 Exercise 5

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

When conducting a two-sample hypothesis test about means with the population standard deviations unknown, we require that the population distributions are **approximately normally distributed** in order to use the t-distribution in the hypothesis test.

Result

Approximately normal

Chapter 12 Exercise 6

Page 460

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

The value of the **test statistic** is determined by using sample information and is used to determine whether to reject the null hypothesis H_0 or not.

Result

Test statistic

Chapter 12 Exercise 7

Page 460

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

In a two-tailed hypothesis test, we need to take into account both tails of the distributions and thus the rejection region will be **evenly split between the two tails**.

Result

Evenly split between the two tails

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

The F-distribution is a continuous probability distribution, which is positively skewed.

Moreover, the F-distribution represents a family of distribution, because different values for the degrees of freedom for the numerator and/or degrees of freedom for the denominator result in a different F-distribution.

However, the F-distribution does not range from $-\infty$ to $+\infty$, because the distribution only ranges from 0 to $+\infty$.

Result

Range from $-\infty$ to $+\infty$

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Two events are **independent**, if the probability that one event occurs in no way affects the probability of the other event occurring.

Two events are **disjoint** or **mutually exclusive**, if the events cannot occur at the same time.

When performing a one-way ANOVA test, we require that the treatments are *independent* such that one treatment won't influence other treatments.

Result

Independent

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of groups} = 4 \\ n_1 &= \text{Sample size} = 6 \\ n_2 &= \text{Sample size} = 6 \\ n_3 &= \text{Sample size} = 6 \\ n_4 &= \text{Sample size} = 6 \\ n &= n_1 + n_2 + n_3 + n_4 = 6 + 6 + 6 + 6 = 24 \end{aligned}$$

The degrees of freedom for the numerator is the number of groups k decreased by 1.

$$d.f._T = k - 1 = 4 - 1 = 3$$

The degrees of freedom for the denominator is the total sample size decreased by the number of groups k.

$$d.f._E = n - k = 24 - 4 = 20$$

Result

3 and 20

Statistics

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PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 18$$
$$\overline{x} = \text{Sample mean} = 96$$
$$s = \text{Sample standard deviation} = 12$$

Given claim: Mean is greater than 90

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 90$$
$$H_a : \mu > 90$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.01$.

$$t = 2.718$$

The rejection region then contains all values greater than 2.718 and thus we reject H_0 when the t-value is larger than 2.718.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{96 - 90}{12/\sqrt{18}} \approx 2.121$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 2.121 < 2.718 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean time in the park is greater than 90 minutes.

Result

Not reasonable

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Given:

$$n_1 = \text{Sample size} = 14$$
$$n_2 = \text{Sample size} = 12$$
$$\overline{x}_1 = \text{Sample mean} = 837$$
$$\overline{x}_2 = \text{Sample mean} = 797$$
$$s_1 = \text{Sample standard deviation} = 30$$
$$s_2 = \text{Sample standard deviation} = 40$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 14 + 12 - 2 = 24$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.064$$

The rejection region then contains all values smaller than -2.064 and all values greater than 2.064,thus we reject H_0 when $t < -2.064$ or $t > 2.064$.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(14 - 1)30^2 + (12 - 1)40^2}{14 + 12 - 2}} \approx 34.9404$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{837 - 797}{34.9404\sqrt{\frac{1}{14} + \frac{1}{12}}} \approx 2.910$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 2.910 > 2.064 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean miles traveled.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

(a) The degrees of freedom for the treatments is given in the row "Between groups" and in the column "df" of the given ANOVA table:

$$d.f._T = 2$$

The degrees of freedom for the treatments is the number of treatments decreased by 1, which implies that the number of treatments is the degrees of freedom for the treatments increased by 1.

$$\text{Number of treatments} = d.f._T + 1 = 2 + 1 = 3$$

Step 2

(b) The total degrees of freedom is given in the row "Total" and in the column "df" of the given ANOVA table:

$$d.f._{total} = 20$$

The total degrees of freedom is the total sample size decreased by 1, which implies that the total sample size is the total degrees of freedom increased by 1.

$$n = \text{Total sample size} = d.f._{total} + 1 = 20 + 1 = 21$$

Step 3

(c) The degrees of freedom for the treatments is given in the row "Between groups" and in the column "df" of the given ANOVA table:

$$d.f._T = 2$$

The degrees of freedom for the error is given in the row "Within groups" and in the column "df" of the given ANOVA table

$$d.f._E = 18$$

The F- critical value is given in the row with $dfd = d.f._E = 18$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.55$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.55.

Step 4

(d) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Step 5

(e) The value of the test statistic is given under "F" in the ANOVA table:

$$F = 4.960047$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 4.960047 > 3.55 \Rightarrow \text{ Reject } H_0$$

Step 6

(e) Since we rejected the null hypothesis in part (e):

There is sufficient evidence to support the claim that the treatment means differ.

Result

(a) 3 treatments

(b) $n = 21$

(c) $F = 3.55$

(d) $H_0 : \mu_1 = \mu_2 = \mu_3, H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$

(e) Reject H_0

(f) Yes